

Assessment 1: Analysing historical data with system performance - Phase 2

Stefan Garevski (33759839)

Part A

1. Analytical Query Design and Implementation

Business Query: "Which DTP regions have the highest rates of serious or fatal road crashes across different times of day, and how do these rates vary across different road types?"

This analysis will look at the severity of road crashes across Victoria by serious or fatal crashes across region, time of day, and road type. Specifically, the analysis will quantify this by looking at the proportion of crashes that are serious or fatal. This provides a more meaningful measure of crash severity because regions or road types with a larger number of crashes are not automatically the locations with the highest severity rate.

The analysis uses the Victorian Road Crash Data dataset which contains approximately 200,000 crash records so this makes the dataset suitable for demonstrating distributed data processing using Spark. The dataset contains the required attributes for this analysis, including ACCIDENT_NO, ACCIDENT_TIME, DTP_REGION, ROAD_TYPE, SERIOUSINJURY, and FATALITY.

The query implements a multi-stage analytical pipeline consisting of several operations.

Time-based analysis: The ACCIDENT_TIME field is used to extract the hour of each crash, and then transformed into four time periods: overnight, morning, afternoon, and evening. This allows the time of day to be analysed rather than treating time as a filter.

Derived severity measure: A SERIOUS_FATAL_CRASH indicator is created. It is assigned a value of 1 when a crash has one or more serious injuries or fatalities, and 0 otherwise. This derived measure is used to calculate the serious/fatal crash rate.

Multi-level aggregation: Records are grouped simultaneously by DTP_REGION, TIME_PERIOD, and ROAD_TYPE. The aggregation calculates total crashes, serious/fatal crashes, serious injuries, and fatalities. This allows comparison between geographic region, time period, and road type.

Post-aggregation filter: Groups with fewer than 100 crashes are removed post-aggregation. This is so that low volume combinations do not produce unstable or misleading severity rates.

Window function: ROW_NUMBER() window function partitions the aggregated results by DTP_REGION and ranks combinations using SERIOUS_FATAL_RATE in descending order. The top three combinations for each region are then retained, which allows comparison of the highest-rate time and road-type combinations within each region.

Just using a simple WHERE filter would only identify individual records satisfying a condition and could not calculate regional severity rates. Similarly, one GROUP BY would produce aggregate statistics but would not identify and rank the highest rate combinations within each region. Therefore, all these operations are necessary in answering the business question.

The query is also appropriate for distributed processing because the dataset contains approximately 200,000 records and the analysis requires aggregation across more than one variable. Spark can distribute the records across partitions, perform aggregations in parallel, shuffle records according to the grouping variables, and perform the final aggregation with window-based ranking.

2. Dataframe Implementation

```
#initial imports
from pyspark.sql import SparkSession
from pyspark.sql.types import *
from pyspark.sql.functions import (
    col,
    when,
    hour,
    to_timestamp,
    countDistinct,
    sum,
    round,
    substring,
    row_number,
    spark_partition_id,
    min,
    max,
    avg,
    count
)
from pyspark.sql.window import Window

#initiate spark session
spark = SparkSession.builder \
    .appName("Victorian Road Crash Analysis") \
    .getOrCreate()

WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-
defaults.properties
26/09/14 20:06:04 WARN Utils: Your hostname, Stefans-MacBook-
Air.local, resolves to a loopback address: 127.0.0.1; using
192.168.1.115 instead (on interface en0)
26/09/14 20:06:04 WARN Utils: Set SPARK_LOCAL_IP if you need to bind
to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-
defaults.properties
Setting default log level to "WARN".
```

To adjust logging level use `sc.setLogLevel(newLevel)`. For SparkR, use `setLogLevel(newLevel)`.
/opt/anaconda3/envs/IT05202/lib/python3.13/site-packages/pyspark/testing/utlis.py:127: FutureWarning: PySpark does not yet fully support pandas >= 3.0.0. Some features may not work correctly. It is recommended to use pandas < 3.0.0 for now.

```
require_minimum_pandas_version()
26/09/14 20:06:05 WARN NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where
applicable
```

#define schema of variables in dataset

```
schema = StructType([
    StructField("ACCIDENT_NO", StringType(), True),
    StructField("ACCIDENT_DATE", StringType(), True),
    StructField("ACCIDENT_TIME", StringType(), True),
    StructField("ACCIDENT_TYPE", StringType(), True),
    StructField("DAY_OF_WEEK", StringType(), True),
    StructField("DCA_CODE", StringType(), True),
    StructField("DCA_CODE_DESCRIPTION", StringType(), True),
    StructField("LIGHT_CONDITION", StringType(), True),
    StructField("POLICE_ATTEND", StringType(), True),
    StructField("ROAD_GEOMETRY", StringType(), True),
    StructField("SEVERITY", StringType(), True),
    StructField("SPEED_ZONE", StringType(), True),
    StructField("RUN_OFFROAD", StringType(), True),
    StructField("ROAD_NAME", StringType(), True),
    StructField("ROAD_TYPE", StringType(), True),
    StructField("ROAD_ROUTE_1", StringType(), True),
    StructField("LGA_NAME", StringType(), True),
    StructField("DTP_REGION", StringType(), True),
    StructField("LATITUDE", DoubleType(), True),
    StructField("LONGITUDE", DoubleType(), True),
    StructField("VICGRID_X", DoubleType(), True),
    StructField("VICGRID_Y", DoubleType(), True),
    StructField("TOTAL_PERSONS", IntegerType(), True),
    StructField("INJ_OR_FATAL", IntegerType(), True),
    StructField("FATALITY", IntegerType(), True),
    StructField("SERIOUSINJURY", IntegerType(), True),
    StructField("OTHERINJURY", IntegerType(), True),
    StructField("NONINJURED", IntegerType(), True),
    StructField("MALES", IntegerType(), True),
    StructField("FEMALES", IntegerType(), True),
    StructField("BICYCLIST", IntegerType(), True),
    StructField("PASSENGER", IntegerType(), True),
    StructField("DRIVER", IntegerType(), True),
    StructField("PEDESTRIAN", IntegerType(), True),
    StructField("PILLION", IntegerType(), True),
    StructField("MOTORCYCLIST", IntegerType(), True),
    StructField("UNKNOWN", IntegerType(), True),
```

```

    StructField("PED_CYCLIST_5_12", IntegerType(), True),
    StructField("PED_CYCLIST_13_18", IntegerType(), True),
    StructField("OLD_PED_65_AND_OVER", IntegerType(), True),
    StructField("OLD_DRIVER_75_AND_OVER", IntegerType(), True),
    StructField("YOUNG_DRIVER_18_25", IntegerType(), True),
    StructField("NO_OF_VEHICLES", IntegerType(), True),
    StructField("HEAVYVEHICLE", IntegerType(), True),
    StructField("PASSENGERVEHICLE", IntegerType(), True),
    StructField("MOTORCYCLE", IntegerType(), True),
    StructField("PT_VEHICLE", IntegerType(), True),
    StructField("DEG_URBAN_NAME", StringType(), True),
    StructField("SRNS", StringType(), True),
    StructField("RMA", StringType(), True),
    StructField("DIVIDED", StringType(), True),
    StructField("STAT_DIV_NAME", StringType(), True)
])

#load in csv data
file_path = "vic_road_crash_data.csv"

crashes = spark.read \
    .option("header", True) \
    .schema(schema) \
    .csv(file_path)

#select relevant columns for business query
crashes_selected = crashes.select(
    "ACCIDENT_NO",
    "ACCIDENT_TIME",
    "DTP_REGION",
    "ROAD_TYPE",
    "SERIOUSINJURY",
    "FATALITY"
)

crashes_selected.show(5)

+-----+-----+-----+-----+-----+
+-----+
| ACCIDENT_NO|ACCIDENT_TIME|    DTP_REGION|ROAD_TYPE|SERIOUSINJURY|
| FATALITY|
+-----+-----+-----+-----+-----+
+-----+
|T20140024624|    18:35:00|    GIPPSLAND|    STREET|    0|
0|
|T20190026336|    15:45:00|    INNER METRO|    ROAD|    0|
0|
|T20190019196|    12:07:00|    INNER METRO|    ROAD|    0|
0|
|T20250029202|    13:10:00|GREATER METRO|    HIGHWAY|    1|

```

```
0|
|T20210005363|      06:30:00|GREATER METRO|      ROAD|      1|
0|
+-----+-----+-----+-----+-----+
+-----+
only showing top 5 rows
```

Create necessary variables to conduct analysis to answer business query

```
#convert accident_time to a single hour number
crashes_time = crashes_selected.withColumn(
  "HOUR",
  substring("ACCIDENT_TIME", 1, 2).cast("int")
)

#categorise the different values of HOUR into different times of day
crashes_time = crashes_time.withColumn(
  "TIME_PERIOD",
  when(col("HOUR") < 6, "Overnight")
  .when(col("HOUR") < 12, "Morning")
  .when(col("HOUR") < 18, "Afternoon")
  .otherwise("Evening")
)

#display top rows to confirm accurate implementation
crashes_time.select(
  "ACCIDENT_TIME",
  "HOUR",
  "TIME_PERIOD"
).show(10)
```

```
+-----+-----+-----+
|ACCIDENT_TIME|HOUR|TIME_PERIOD|
+-----+-----+-----+
|      18:35:00|  18|    Evening|
|      15:45:00|  15|   Afternoon|
|      12:07:00|  12|   Afternoon|
|      13:10:00|  13|   Afternoon|
|      06:30:00|   6|    Morning|
|      16:55:00|  16|   Afternoon|
|      13:28:00|  13|   Afternoon|
|      13:30:00|  13|   Afternoon|
|      10:20:00|  10|    Morning|
|      15:00:00|  15|   Afternoon|
+-----+-----+-----+
only showing top 10 rows
```

```
#create indicator of whether the accident resulted in 1 or more
serious injuries or fatality
```

```

crashes_prepared = crashes_time.withColumn(
  "SERIOUS_FATAL_CRASH",
  when(
    (col("SERIOUSINJURY") > 0) | (col("FATALITY") > 0),
    1
  ).otherwise(0)
)

#display
crashes_prepared.select(
  "ACCIDENT_NO",
  "SERIOUSINJURY",
  "FATALITY",
  "SERIOUS_FATAL_CRASH"
).show(10)

```

ACCIDENT_NO	SERIOUSINJURY	FATALITY	SERIOUS_FATAL_CRASH
T20140024624	0	0	0
T20190026336	0	0	0
T20190019196	0	0	0
T20250029202	1	0	1
T20210005363	1	0	1
T20210014188	0	0	0
T20250004689	0	0	0
T20200020837	1	0	1
T20220026535	0	0	0
T20240003419	0	0	0

only showing top 10 rows

Cache data at this point

The dataframe `crashes_prepared` should be cached at this point because it is the dataset that will be used for aggregation and analysis. Caching will avoid needing to recalculate the generated fields such as `time_period` and `serious_fatal_crash` if the dataframe is reused. Aggregate results that are generated subsequently do not need to be cached as their computation is a lot smaller. For example the count statement that follows calls upon the cache, as is able to be processed quickly.

```

crashes_prepared.cache()

# Trigger the cache
crashes_prepared.count()

```

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Aggregation

```
#record crashes by DTP region, time period, and road type
aggregated = crashes_prepared.groupBy(
    "DTP_REGION",
    "TIME_PERIOD",
    "ROAD_TYPE"
).agg(
    countDistinct("ACCIDENT_NO").alias("TOTAL_CRASHES"),
    sum("SERIOUS_FATAL_CRASH").alias("SERIOUS_FATAL_CRASHES"),
    sum("SERIOUS_INJURY").alias("SERIOUS_INJURIES"),
    sum("FATALITY").alias("FATALITIES")
)

aggregated.show(10)
```

```
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      DTP_REGION|TIME_PERIOD|ROAD_TYPE|TOTAL_CRASHES|
|SERIOUS_FATAL_CRASHES|SERIOUS_INJURIES|FATALITIES|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      INNER METRO|Morning|ESPLANADE|55|
24|25|1|
|GREATER METRO|Evening|HIGHWAY|1519|
598|736|28|
|GREATER METRO|Overnight|LANE|8|
4|4|0|
|BARWON SOUTH WEST|Evening|RAMP|12|
6|8|0|
|LODDON MALLEE|Overnight|PLACE|1|
1|1|0|
|INNER METRO|Afternoon|NULL|539|
183|208|3|
|HUME|Evening|NULL|18|
8|15|0|
|GRAMPIANS|Evening|ROAD|765|
407|469|35|
|GREATER METRO|Afternoon|GROVE|52|
23|25|0|
|LODDON MALLEE|Afternoon|ESPLANADE|2|
0|0|0|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
only showing top 10 rows
```

Calculate serious/fatal crash rate

```
#create ratio using number of crashes resulting in injury or death
over total number of crashes
```

```

results = aggregated.withColumn(
    "SERIOUS_FATAL_RATE",
    round(
        col("SERIOUS_FATAL_CRASHES") /
        col("TOTAL_CRASHES") * 100,
        2
    )
)

results.show(10)

```

```

+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      DTP_REGION|TIME_PERIOD|ROAD_TYPE|TOTAL_CRASHES|
SERIOUS_FATAL_CRASHES|SERIOUS_INJURIES|FATALITIES|SERIOUS_FATAL_RATE|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      INNER METRO|Morning|ESPLANADE|55|
24|25|1|43.64|
|GREATER METRO|Evening|HIGHWAY|1519|
598|736|28|39.37|
|GREATER METRO|Overnight|LANE|8|
4|4|0|50.0|
|BARWON SOUTH WEST|Evening|RAMP|12|
6|8|0|50.0|
|LODDON MALLEE|Overnight|PLACE|1|
1|1|0|100.0|
|INNER METRO|Afternoon|NULL|539|
183|208|3|33.95|
|HUME|Evening|NULL|18|
8|15|0|44.44|
|GRAMPIANS|Evening|ROAD|765|
407|469|35|53.2|
|GREATER METRO|Afternoon|GROVE|52|
23|25|0|44.23|
|LODDON MALLEE|Afternoon|ESPLANADE|2|
0|0|0|0.0|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
only showing top 10 rows

```

Remove groups that contain less than 100 crashes

```

filtered_results = results.filter(
    col("TOTAL_CRASHES") >= 100
)

filtered_results.show(10)

```

```

+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      DTP_REGION|TIME_PERIOD|ROAD_TYPE|TOTAL_CRASHES|
SERIOUS_FATAL_CRASHES|SERIOUS_INJURIES|FATALITIES|SERIOUS_FATAL_RATE|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
|    GREATER METRO|    Evening|    HIGHWAY|    1519|
598|          736|        28|    39.37|
|    INNER METRO|    Afternoon|    NULL|    539|
183|          208|        3|    33.95|
|    GRAMPIANS|    Evening|    ROAD|    765|
407|          469|       35|    53.2|
|          HUME|    Afternoon|    HIGHWAY|    699|
356|          452|       49|    50.93|
|BARWON SOUTH WEST|    Afternoon|    DRIVE|    104|
46|          48|        0|    44.23|
|    GIPPSLAND|    Evening|    STREET|    341|
98|          104|        4|    28.74|
|BARWON SOUTH WEST|    Overnight|    ROAD|    448|
271|          295|       31|    60.49|
|    INNER METRO|    Morning|    CRESCENT|    127|
43|          44|        0|    33.86|
|    GREATER METRO|    Morning|    FREEWAY|    1058|
425|          475|       15|    40.17|
|    INNER METRO|    Afternoon|    HIGHWAY|    1135|
368|          406|       11|    32.42|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
only showing top 10 rows

```

Rank DTP + ROAD_TYPE in descending serious/fatal crash rate

```

#create window and partition data by DTP region
window_spec = Window \
    .partitionBy("DTP_REGION") \
    .orderBy(col("SERIOUS_FATAL_RATE").desc())

#create field that ranks DTP and road_type by the crash rate
ranked_results = filtered_results.withColumn(
    "REGION_RANK",
    row_number().over(window_spec)
)

ranked_results.show(10)

```

```

+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
|      DTP_REGION|TIME_PERIOD|ROAD_TYPE|TOTAL_CRASHES|
SERIOUS_FATAL_CRASHES|SERIOUS_INJURIES|FATALITIES|SERIOUS_FATAL_RATE|

```

```

REGION_RANK|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
|BARWON SOUTH WEST| Overnight| HIGHWAY| 154|
103| 109| 11| 66.88| 1|
|BARWON SOUTH WEST| Overnight| ROAD| 448|
271| 295| 31| 60.49| 2|
|BARWON SOUTH WEST| Evening| HIGHWAY| 389|
218| 281| 20| 56.04| 3|
|BARWON SOUTH WEST| Evening| ROAD| 1274|
704| 907| 41| 55.26| 4|
|BARWON SOUTH WEST| Morning| ROAD| 2051|
1045| 1227| 58| 50.95| 5|
|BARWON SOUTH WEST| Overnight| STREET| 162|
82| 92| 4| 50.62| 6|
|BARWON SOUTH WEST| Afternoon| ROAD| 3075|
1528| 1929| 85| 49.69| 7|
|BARWON SOUTH WEST| Afternoon| AVENUE| 209|
101| 107| 5| 48.33| 8|
|BARWON SOUTH WEST| Morning| HIGHWAY| 577|
277| 338| 22| 48.01| 9|
|BARWON SOUTH WEST| Afternoon| HIGHWAY| 828|
391| 509| 33| 47.22| 10|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
only showing top 10 rows

```

Generate top 3 worst road types at a particular time period when considering the serious/fatal crash rate

```

#filter for top 3 from each region
top_results = ranked_results.filter(
  col("REGION_RANK") <= 3
)

#generate summary output
final_results = top_results.select(
  "DTP_REGION",
  "TIME_PERIOD",
  "ROAD_TYPE",
  "TOTAL_CRASHES",
  "SERIOUS_FATAL_CRASHES",
  "SERIOUS_INJURIES",
  "FATALITIES",
  "SERIOUS_FATAL_RATE",
  "REGION_RANK"
).orderBy(

```

```

    "DTP_REGION",
    "REGION_RANK"
)

#show output
final_results.show(25)

+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
|      DTP_REGION|TIME_PERIOD|ROAD_TYPE|TOTAL_CRASHES|
SERIOUS_FATAL_CRASHES|SERIOUS_INJURIES|FATALITIES|SERIOUS_FATAL_RATE|
REGION_RANK|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
|BARWON SOUTH WEST| Overnight| HIGHWAY|      154|
103|           109|      11|      66.88|      1|
|BARWON SOUTH WEST| Overnight|   ROAD|      448|
271|           295|      31|      60.49|      2|
|BARWON SOUTH WEST| Evening| HIGHWAY|      389|
218|           281|      20|      56.04|      3|
|      GIPPSLAND| Overnight|   ROAD|      370|
190|           196|      28|      51.35|      1|
|      GIPPSLAND| Overnight| HIGHWAY|      113|
53|           51|      11|      46.9|      2|
|      GIPPSLAND| Evening|   ROAD|      930|
414|           452|      48|      44.52|      3|
|      GRAMPIANS| Overnight|   ROAD|      243|
135|           137|      21|      55.56|      1|
|      GRAMPIANS| Overnight| HIGHWAY|      126|
69|           76|      11|      54.76|      2|
|      GRAMPIANS| Afternoon| HIGHWAY|      546|
292|           390|      44|      53.48|      3|
| GREATER METRO| Overnight| HIGHWAY|      427|
225|           256|      12|      52.69|      1|
| GREATER METRO| Overnight| FREEWAY|      288|
146|           149|      14|      50.69|      2|
| GREATER METRO| Overnight|   ROAD|      2240|
1105|           1229|     104|      49.33|      3|
|           HUME| Overnight| HIGHWAY|      118|
70|           70|      11|      59.32|      1|
|           HUME| Overnight|   ROAD|      436|
236|           250|      34|      54.13|      2|
|           HUME| Afternoon| FREEWAY|      306|
159|           208|      21|      51.96|      3|
|      INNER METRO| Overnight| HIGHWAY|      133|
85|           87|       9|      63.91|      1|
|      INNER METRO| Overnight|   ROAD|      2063|
1070|           1208|      55|      51.87|      2|

```

	INNER METRO	Overnight	FREEWAY	217	
99	121	6	45.62	3	
	LODDON MALLEE	Overnight	HIGHWAY	141	
74	72	17	52.48	1	
	LODDON MALLEE	Overnight	ROAD	329	
171	181	21	51.98	2	
	LODDON MALLEE	Evening	ROAD	894	
441	483	44	49.33	3	
+-----+-----+-----+-----+					
+-----+-----+-----+-----+					
+-----+					

View execution plan

```
final_results.explain("formatted")
```

```
== Physical Plan ==
```

```
AdaptiveSparkPlan (23)
```

```
+ - Sort (22)
```

```
  + - Exchange (21)
```

```
    + - Filter (20)
```

```
      + - Window (19)
```

```
        + - WindowGroupLimit (18)
```

```
          + - Sort (17)
```

```
            + - Exchange (16)
```

```
              + - WindowGroupLimit (15)
```

```
                + - Sort (14)
```

```
                  + - Project (13)
```

```
                    + - Filter (12)
```

```
                      + - HashAggregate (11)
```

```
                        + - Exchange (10)
```

```
                          + - HashAggregate (9)
```

```
                            + - HashAggregate (8)
```

```
                              + - Exchange (7)
```

```
                                + - HashAggregate (6)
```

```
                                  + -
```

```
InMemoryTableScan (1)
```

```
                                + -
```

```
InMemoryRelation (2)
```

```
                                + - *
```

```
Project (5)
```

```
                                + - *
```

```
Project (4)
```

```
+ - Scan csv (3)
```

```
(1) InMemoryTableScan
```

```
Output [7]: [ACCIDENT_NO#0, DTP_REGION#17, ROAD_TYPE#14,
```

SERIOUSINJURY#25, FATALITY#24, TIME_PERIOD#79, SERIOUS_FATAL_CRASH#91]
Arguments: [ACCIDENT_NO#0, DTP_REGION#17, ROAD_TYPE#14,
SERIOUSINJURY#25, FATALITY#24, TIME_PERIOD#79, SERIOUS_FATAL_CRASH#91]

(2) InMemoryRelation

Arguments: [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78, TIME_PERIOD#79,
SERIOUS_FATAL_CRASH#91], StorageLevel(disk, memory, deserialized, 1
replicas)

(3) Scan csv

Output [6]: [ACCIDENT_NO#0, ACCIDENT_TIME#2, ROAD_TYPE#14,
DTP_REGION#17, FATALITY#24, SERIOUSINJURY#25]

Batched: false

Location: InMemoryFileIndex [file:/Users/stefangarevski/Desktop/Data
Processing for Big Data IT05202/Data Processing for Big Data
IT05202/Assessment 1/vic_road_crash_data.csv]

ReadSchema:

struct<ACCIDENT_NO:string,ACCIDENT_TIME:string,ROAD_TYPE:string,DTP_RE
GION:string,FATALITY:int,SERIOUSINJURY:int>

(4) Project [codegen id : 1]

Output [7]: [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24,
cast(substring(ACCIDENT_TIME#2, 1, 2) as int) AS HOUR#78]

Input [6]: [ACCIDENT_NO#0, ACCIDENT_TIME#2, ROAD_TYPE#14,
DTP_REGION#17, FATALITY#24, SERIOUSINJURY#25]

(5) Project [codegen id : 1]

Output [9]: [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78, CASE WHEN
(HOUR#78 < 6) THEN Overnight WHEN (HOUR#78 < 12) THEN Morning WHEN
(HOUR#78 < 18) THEN Afternoon ELSE Evening END AS TIME_PERIOD#79, CASE
WHEN ((SERIOUSINJURY#25 > 0) OR (FATALITY#24 > 0)) THEN 1 ELSE 0 END
AS SERIOUS_FATAL_CRASH#91]

Input [7]: [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78]

(6) HashAggregate

Input [7]: [ACCIDENT_NO#0, DTP_REGION#17, ROAD_TYPE#14,
SERIOUSINJURY#25, FATALITY#24, TIME_PERIOD#79, SERIOUS_FATAL_CRASH#91]

Keys [4]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0]

Functions [3]: [partial_sum(SERIOUS_FATAL_CRASH#91),
partial_sum(SERIOUSINJURY#25), partial_sum(FATALITY#24)]

Aggregate Attributes [3]: [sum(SERIOUS_FATAL_CRASH#91)#411L,
sum(SERIOUSINJURY#25)#412L, sum(FATALITY#24)#413L]

Results [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14,
ACCIDENT_NO#0, sum#557L, sum#559L, sum#561L]

(7) Exchange

Input [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0, sum#557L, sum#559L, sum#561L]
Arguments: hashpartitioning(DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0, 200), ENSURE_REQUIREMENTS, [plan_id=1281]

(8) HashAggregate

Input [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0, sum#557L, sum#559L, sum#561L]
Keys [4]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0]
Functions [3]: [merge_sum(SERIOUS_FATAL_CRASH#91), merge_sum(SERIOUSINJURY#25), merge_sum(FATALITY#24)]
Aggregate Attributes [3]: [sum(SERIOUS_FATAL_CRASH#91)#411L, sum(SERIOUSINJURY#25)#412L, sum(FATALITY#24)#413L]
Results [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0, sum#557L, sum#559L, sum#561L]

(9) HashAggregate

Input [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0, sum#557L, sum#559L, sum#561L]
Keys [3]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14]
Functions [4]: [merge_sum(SERIOUS_FATAL_CRASH#91), merge_sum(SERIOUSINJURY#25), merge_sum(FATALITY#24), partial_count(distinct ACCIDENT_NO#0)]
Aggregate Attributes [4]: [sum(SERIOUS_FATAL_CRASH#91)#411L, sum(SERIOUSINJURY#25)#412L, sum(FATALITY#24)#413L, count(ACCIDENT_NO#0)#410L]
Results [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, sum#557L, sum#559L, sum#561L, count#564L]

(10) Exchange

Input [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, sum#557L, sum#559L, sum#561L, count#564L]
Arguments: hashpartitioning(DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, 200), ENSURE_REQUIREMENTS, [plan_id=1285]

(11) HashAggregate

Input [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, sum#557L, sum#559L, sum#561L, count#564L]
Keys [3]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14]
Functions [4]: [sum(SERIOUS_FATAL_CRASH#91), sum(SERIOUSINJURY#25), sum(FATALITY#24), count(distinct ACCIDENT_NO#0)]
Aggregate Attributes [4]: [sum(SERIOUS_FATAL_CRASH#91)#411L, sum(SERIOUSINJURY#25)#412L, sum(FATALITY#24)#413L, count(ACCIDENT_NO#0)#410L]
Results [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, count(ACCIDENT_NO#0)#410L AS TOTAL_CRASHES#397L, sum(SERIOUS_FATAL_CRASH#91)#411L AS SERIOUS_FATAL_CRASHES#398L, sum(SERIOUSINJURY#25)#412L AS SERIOUS_INJURIES#399L, sum(FATALITY#24)#413L AS FATALITIES#400L]

(12) Filter

Input [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L, FATALITIES#400L]

Condition : (TOTAL_CRASHES#397L >= 100)

(13) Project

Output [8]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L, FATALITIES#400L, round(((cast(SERIOUS_FATAL_CRASHES#398L as double) / cast(TOTAL_CRASHES#397L as double)) * 100.0), 2) AS

SERIOUS_FATAL_RATE#710]

Input [7]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L, FATALITIES#400L]

(14) Sort

Input [8]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L, FATALITIES#400L, SERIOUS_FATAL_RATE#710]

Arguments: [DTP_REGION#17 ASC NULLS FIRST, SERIOUS_FATAL_RATE#710 DESC NULLS LAST], false, 0

(15) WindowGroupLimit

Input [8]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L, FATALITIES#400L, SERIOUS_FATAL_RATE#710]

Arguments: [DTP_REGION#17], [SERIOUS_FATAL_RATE#710 DESC NULLS LAST], row_number(), 3, Partial

(16) Exchange

Input [8]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L, FATALITIES#400L, SERIOUS_FATAL_RATE#710]

Arguments: hashpartitioning(DTP_REGION#17, 200), ENSURE_REQUIREMENTS, [plan_id=1293]

(17) Sort

Input [8]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L, FATALITIES#400L, SERIOUS_FATAL_RATE#710]

Arguments: [DTP_REGION#17 ASC NULLS FIRST, SERIOUS_FATAL_RATE#710 DESC NULLS LAST], false, 0

(18) WindowGroupLimit

Input [8]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L, FATALITIES#400L, SERIOUS_FATAL_RATE#710]

Arguments: [DTP_REGION#17], [SERIOUS_FATAL_RATE#710 DESC NULLS LAST], row_number(), 3, Final

```

(19) Window
Input [8]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14,
TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L,
FATALITIES#400L, SERIOUS_FATAL_RATE#710]
Arguments: [row_number() window specification(DTP_REGION#17,
SERIOUS_FATAL_RATE#710 DESC NULLS LAST, specified window frame(RowFrame,
unbounded preceding $(), current row $())) AS REGION_RANK#1293],
[DTP_REGION#17], [SERIOUS_FATAL_RATE#710 DESC NULLS LAST]

(20) Filter
Input [9]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14,
TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L,
FATALITIES#400L, SERIOUS_FATAL_RATE#710, REGION_RANK#1293]
Condition : (REGION_RANK#1293 <= 3)

(21) Exchange
Input [9]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14,
TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L,
FATALITIES#400L, SERIOUS_FATAL_RATE#710, REGION_RANK#1293]
Arguments: range partitioning(DTP_REGION#17 ASC NULLS FIRST,
REGION_RANK#1293 ASC NULLS FIRST, 200), ENSURE_REQUIREMENTS,
[plan_id=1299]

(22) Sort
Input [9]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14,
TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L,
FATALITIES#400L, SERIOUS_FATAL_RATE#710, REGION_RANK#1293]
Arguments: [DTP_REGION#17 ASC NULLS FIRST, REGION_RANK#1293 ASC NULLS
FIRST], true, 0

(23) AdaptiveSparkPlan
Output [9]: [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14,
TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L,
FATALITIES#400L, SERIOUS_FATAL_RATE#710, REGION_RANK#1293]
Arguments: isFinalPlan=false

```

Optimisation discussion

It was important to use `select()` as a way to project in this instance because there are a lot of variables in the dataset that are not necessary to the analysis required to answer the business query. This demonstrates Catalyst optimisation as the `select` avoids carrying unnecessary variables from the CSV, only selecting the ones that are required for aggregation.

The variables 1. serious/fatal crash indicator and 2. the time period transformation are done prior to aggregation, whereas filtering for a small volume of crashes is done after aggregation from the smaller aggregated dataset. This means that predicate pushdown is limited in this query as the filtering was done post aggregation. Caching is done on the dataset that contains all data

required for the business query analysis. No joining is necessary in this instance as all necessary variables are present in the Victorian Crash Data table. This is good as joins would introduce potentially unnecessary data movement. Spark determines an efficient way to execute all these tasks so that there is very little unnecessary data movement and aggregation is done in an optimised way.

3. Spark SQL Implementation

Create temp view of data

```
#create temp view
crashes_selected.createOrReplaceTempView("crashes")

#verify view
spark.sql("SELECT * FROM crashes LIMIT 5").show()
```

ACCIDENT_NO	ACCIDENT_TIME	DTP_REGION	ROAD_TYPE	SERIOUSINJURY	FATALITY
T20140024624	18:35:00	GIPPSLAND	STREET	0	0
T20190026336	15:45:00	INNER METRO	ROAD	0	0
T20190019196	12:07:00	INNER METRO	ROAD	0	0
T20250029202	13:10:00	GREATER METRO	HIGHWAY	1	0
T20210005363	06:30:00	GREATER METRO	ROAD	1	0

26/09/14 20:06:46 WARN SparkStringUtils: Truncated the string representation of a plan since it was too large. This behavior can be adjusted by setting 'spark.sql.debug.maxToStringFields'.

SQL query to answer business query

```
#time_data CTE creates HOUR to eventually create TIME_PERIOD
#prepared_data creates TIME_PERIOD and SERIOUS_FATAL_CRASH flags
#aggregated_data creates summary data including SERIOUS_FATAL_RATE,
and filters for COUNT(ACCIDENT_NO) >= 100
#ranked_data orders the DTP_REGION by SERIOUS_FATAL_RATE and creates
rank
#final select presents only top 3 TIME_PERIOD, ROAD_TYPE for each
DTP_REGION in descending order
```

```

sql_query = """
WITH time_data AS (
    SELECT
        ACCIDENT_NO,
        DTP_REGION,
        ROAD_TYPE,
        SERIOUSINJURY,
        FATALITY,
        CAST(SUBSTRING(ACCIDENT_TIME, 1, 2) AS INT) AS HOUR
    FROM crashes
),
prepared_data AS (
    SELECT
        ACCIDENT_NO,
        DTP_REGION,
        ROAD_TYPE,
        SERIOUSINJURY,
        FATALITY,
        CASE
            WHEN HOUR < 6 THEN 'Overnight'
            WHEN HOUR < 12 THEN 'Morning'
            WHEN HOUR < 18 THEN 'Afternoon'
            ELSE 'Evening'
        END AS TIME_PERIOD,
        CASE
            WHEN SERIOUSINJURY > 0 OR FATALITY > 0 THEN 1
            ELSE 0
        END AS SERIOUS_FATAL_CRASH
    FROM time_data
),
aggregated_data AS (
    SELECT
        DTP_REGION,
        TIME_PERIOD,
        ROAD_TYPE,
        COUNT(ACCIDENT_NO) AS TOTAL_CRASHES,
        SUM(SERIOUS_FATAL_CRASH) AS SERIOUS_FATAL_CRASHES,
        SUM(SERIOUSINJURY) AS SERIOUS_INJURIES,
        SUM(FATALITY) AS FATALITIES,
        ROUND(
            SUM(SERIOUS_FATAL_CRASH) / COUNT(ACCIDENT_NO) * 100,
            2
        ) AS SERIOUS_FATAL_RATE
    FROM prepared_data
    GROUP BY
        DTP_REGION,
        TIME_PERIOD,

```

```

        ROAD_TYPE
    HAVING COUNT(ACCIDENT_NO) >= 100
),
ranked_data AS (
    SELECT
        DTP_REGION,
        TIME_PERIOD,
        ROAD_TYPE,
        TOTAL_CRASHES,
        SERIOUS_FATAL_CRASHES,
        SERIOUS_INJURIES,
        FATALITIES,
        SERIOUS_FATAL_RATE,
        ROW_NUMBER() OVER (
            PARTITION BY DTP_REGION
            ORDER BY SERIOUS_FATAL_RATE DESC
        ) AS REGION_RANK
    FROM aggregated_data
)

```

```

SELECT *
FROM ranked_data
WHERE REGION_RANK <= 3
ORDER BY
    DTP_REGION,
    REGION_RANK
"""

```

```
sql_results = spark.sql(sql_query)
```

```
sql_results.show(25)
```

```

+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
|      DTP_REGION|TIME_PERIOD|ROAD_TYPE|TOTAL_CRASHES|
SERIOUS_FATAL_CRASHES|SERIOUS_INJURIES|FATALITIES|SERIOUS_FATAL_RATE|
REGION_RANK|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
|BARWON SOUTH WEST| Overnight| HIGHWAY|      154|
103|      109|      11|      66.88|      1|
|BARWON SOUTH WEST| Overnight|   ROAD|      448|
271|      295|      31|      60.49|      2|
|BARWON SOUTH WEST|   Evening| HIGHWAY|      389|
218|      281|      20|      56.04|      3|
|      GIPPSLAND| Overnight|   ROAD|      370|
190|      196|      28|      51.35|      1|

```

	GIPPSLAND	Overnight	HIGHWAY	113	
53	51	11	46.9	2	
	GIPPSLAND	Evening	ROAD	930	
414	452	48	44.52	3	
	GRAMPIANS	Overnight	ROAD	243	
135	137	21	55.56	1	
	GRAMPIANS	Overnight	HIGHWAY	126	
69	76	11	54.76	2	
	GRAMPIANS	Afternoon	HIGHWAY	546	
292	390	44	53.48	3	
	GREATER METRO	Overnight	HIGHWAY	427	
225	256	12	52.69	1	
	GREATER METRO	Overnight	FREEWAY	288	
146	149	14	50.69	2	
	GREATER METRO	Overnight	ROAD	2240	
1105	1229	104	49.33	3	
	HUME	Overnight	HIGHWAY	118	
70	70	11	59.32	1	
	HUME	Overnight	ROAD	436	
236	250	34	54.13	2	
	HUME	Afternoon	FREEWAY	306	
159	208	21	51.96	3	
	INNER METRO	Overnight	HIGHWAY	133	
85	87	9	63.91	1	
	INNER METRO	Overnight	ROAD	2063	
1070	1208	55	51.87	2	
	INNER METRO	Overnight	FREEWAY	217	
99	121	6	45.62	3	
	LODDON MALLEE	Overnight	HIGHWAY	141	
74	72	17	52.48	1	
	LODDON MALLEE	Overnight	ROAD	329	
171	181	21	51.98	2	
	LODDON MALLEE	Evening	ROAD	894	
441	483	44	49.33	3	
+-----+-----+-----+-----+-----					
+-----+-----+-----+-----+-----					
+-----+					

DataFrame and Spark SQL differences discussion (readability, maintainability, query structure)

The spark dataframe and spark SQL codes produce the same results as they perform the same operations. The dataframe implementation might be considered easier to follow as the commands are more closely linked to the data, such as `groupBy()` and `agg()`. The spark SQL implementation is more structured with the use of CTEs to generate and complete different parts of the business query analysis (one CTE to create the required feature, another to aggregate the data, another to rank, etc.). Both queries are maintainable, the dataframe one if the analyst is

more familiar with python, and the SQL one is better if the operations required are more relational (select, join, etc.).

The SQL implementation structures the operations through the CTEs `time_data`, `prepared_data`, `aggregated_data`, and `ranked_data`, so it makes the process from raw data to final results clear and aligned to the business query. The dataframe does the same using chained python transformations, which could provide more flexibility if the transformations need to be constructed dynamically.

For debuggability, you could argue the SQL implementation is better since the CTEs allow for easier identification of whether the hour extraction is appropriate by inspecting `prepared_data`.

Overall, the SQL approach is good since the analysis calls for mainly relational operations, while the dataframe integrates more reusable functions for better python-based processing that might be required.

4. Result Validation

Compare number of results

```
print("DataFrame API rows:", final_results.count())
print("Spark SQL rows:", sql_results.count())
```

```
DataFrame API rows: 21
Spark SQL rows: 21
```

Compare results

```
df_api = final_results.orderBy(
    "DTP_REGION",
    "REGION_RANK"
)
```

```
df_sql = sql_results.orderBy(
    "DTP_REGION",
    "REGION_RANK"
)
```

```
df_api.show(21)
df_sql.show(21)
```

```
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
|      DTP_REGION|TIME_PERIOD|ROAD_TYPE|TOTAL_CRASHES|
SERIOUS_FATAL_CRASHES|SERIOUS_INJURIES|FATALITIES|SERIOUS_FATAL_RATE|
REGION_RANK|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+
|BARWON SOUTH WEST| Overnight| HIGHWAY|          154|
```


SERIOUS_FATAL_CRASHES|SERIOUS_INJURIES|FATALITIES|SERIOUS_FATAL_RATE|
REGION_RANK|[illegible]

```
+-----+
```

Discuss differences in results

There are no differences in results between the dataframes and SQL implementations.

Both implementations perform the same sequence of operations with extraction of the accident hour, classification into time periods, creation of the serious/fatal crash indicator, aggregation by DTP_REGION, TIME_PERIOD and ROAD_TYPE, post-aggregation filtering, calculation of the serious/fatal crash rate, and ranking within each region. The equivalent input data and transformation logic should produce equivalent results.

It is important to make sure the ORDER BY is applied at an equivalent point in the query as this could produce different results, but that was already done in the above queries. Same for numerical formatting; this could produce different results based on how the operations are applied as there could be differences in how division and decimal point rounding is applied by each of the implementations. Both of these instances are relatively minor differences if they were present, it is more about how the data output is presented.

Part B: System perspective and performance analysis

1. Partition Strategy

```
#start with partition of 10  
n = 10
```

1.1 Hash Partition

```
#hash partition function  
hash_df = crashes.repartition(n, "ACCIDENT_NO")  
  
#create summary of count in each partition  
hash_partition_counts = hash_df \  
    .withColumn("PARTITION_ID", spark_partition_id()) \  
    .groupBy("PARTITION_ID") \  
    .agg(count("*").alias("RECORD_COUNT")) \  
    .orderBy("PARTITION_ID")  
  
#display partition and number of records in each  
hash_partition_counts.show()
```

```
+-----+-----+  
|PARTITION_ID|RECORD_COUNT|  
+-----+-----+  
|          0|         19910|
```

1	20005
2	19876
3	19903
4	19990
5	20312
6	20065
7	20206
8	20150
9	19935

1.2 Range Partition

```
#partition by range method
range_df = crashes.repartitionByRange(n, "ACCIDENT_NO")

#create summary of count in each partition
range_partition_counts = range_df \
    .withColumn("PARTITION_ID", spark_partition_id()) \
    .groupBy("PARTITION_ID") \
    .agg(count("*").alias("RECORD_COUNT")) \
    .orderBy("PARTITION_ID")

#display partition and number of records in each
range_partition_counts.show()
```

PARTITION_ID	RECORD_COUNT
0	19887
1	19927
2	19206
3	20467
4	19517
5	18158
6	21410
7	20064
8	20050
9	21666

1.3 Comparative Analysis

```
#rename hash partition count table
hash_counts = hash_partition_counts \
    .withColumnRenamed("RECORD_COUNT", "HASH_RECORD_COUNT")

#rename range partition count table
```

```

range_counts = range_partition_counts \
    .withColumnRenamed("RECORD_COUNT", "RANGE_RECORD_COUNT")

#join volumes in each partition between partition method
partition_comparison = hash_counts.join(
    range_counts,
    on="PARTITION_ID",
    how="outer"
).orderBy("PARTITION_ID")

#show comparison table
partition_comparison.show()

#compute general stats for hash partition
hash_stats = hash_partition_counts.agg(
    min("RECORD_COUNT").alias("MIN_RECORDS"),
    max("RECORD_COUNT").alias("MAX_RECORDS"),
    avg("RECORD_COUNT").alias("AVG_RECORDS")
)

#compute general stats for range partition
range_stats = range_partition_counts.agg(
    min("RECORD_COUNT").alias("MIN_RECORDS"),
    max("RECORD_COUNT").alias("MAX_RECORDS"),
    avg("RECORD_COUNT").alias("AVG_RECORDS")
)

print("Hash partition statistics:")
hash_stats.show()

print("Range partition statistics:")
range_stats.show()

```

PARTITION_ID	HASH_RECORD_COUNT	RANGE_RECORD_COUNT
0	19910	18334
1	20005	19717
2	19876	19444
3	19903	21960
4	19990	19847
5	20312	18855
6	20065	21847
7	20206	20558
8	20150	19820
9	19935	19970

Hash partition statistics:

```

+-----+-----+-----+

```

MIN_RECORDS	MAX_RECORDS	AVG_RECORDS
19876	20312	20035.2

Range partition statistics:

MIN_RECORDS	MAX_RECORDS	AVG_RECORDS
19008	21420	20035.2

Hash and range partitioning were completed using the high cardinality variable ACCIDENT_NO, and 10 partitions were created. This means that each partition should have around 20,000 observations present, which is a good balance to observe the distribution of data in each one and is not intense when running using spark.

From the summary statistics, it appears that hash partitioning is more even, with the difference between the min_records and max_records being more narrow than the range partition statistics. Both methods produce the same average of 20035 records though, and this is because the number of total rows in the data and the number of partitions were consistent between the methods.

The hash partitioning shows very little evidence of data skew since the partition sizes are very similar to one another, while the range partition gives more observations to some partitions compared to others. The range is not extremely different so this probably isn't evident of severe skew. This is because ACCIDENT_NO values might not be evenly distributed across the ranges selected by the range partitioning. If there was skew present through another partitioning strategy, hash partitioning would be a suitable mitigation method in this instance.

In this instance, the hash partitioning is more suitable as the result is more even and gives opportunity to more even workload across the partitions. This will likely lead to better query performance than compared to range partitioning. On the other hand, if there is a need for ordered operations, that is when range partitioning might be more useful, but it is still a little more uneven.

2. Execution time benchmarking

Dataframe benchmarking

```
%%time

benchmark_df = crashes_selected \
    .withColumn(
        "HOUR",
        substring("ACCIDENT_TIME", 1, 2).cast("int")
    ) \
    .withColumn(
        "TIME_PERIOD",
```

```

        when(col("HOUR") < 6, "Overnight")
        .when(col("HOUR") < 12, "Morning")
        .when(col("HOUR") < 18, "Afternoon")
        .otherwise("Evening")
    ) \
    .withColumn(
        "SERIOUS_FATAL_CRASH",
        when(
            (col("SERIOUSINJURY") > 0) | (col("FATALITY") > 0),
            1
        ).otherwise(0)
    ) \
    .groupBy(
        "DTP_REGION",
        "TIME_PERIOD",
        "ROAD_TYPE"
    ) \
    .agg(
        count("ACCIDENT_NO").alias("TOTAL_CRASHES"),
        sum("SERIOUS_FATAL_CRASH").alias("SERIOUS_FATAL_CRASHES"),
        sum("SERIOUSINJURY").alias("SERIOUS_INJURIES"),
        sum("FATALITY").alias("FATALITIES")
    ) \
    .withColumn(
        "SERIOUS_FATAL_RATE",
        round(
            col("SERIOUS_FATAL_CRASHES") /
            col("TOTAL_CRASHES") * 100,
            2
        )
    ) \
    .filter(col("TOTAL_CRASHES") >= 100)

window_spec = Window \
    .partitionBy("DTP_REGION") \
    .orderBy(col("SERIOUS_FATAL_RATE").desc())

benchmark_df = benchmark_df.withColumn(
    "REGION_RANK",
    row_number().over(window_spec)
)

benchmark_df = benchmark_df \
    .filter(col("REGION_RANK") <= 3) \
    .orderBy("DTP_REGION", "REGION_RANK")

benchmark_df.collect()

CPU times: user 11.2 ms, sys: 5.86 ms, total: 17.1 ms
Wall time: 324 ms

```

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[Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=103,
SERIOUS_INJURIES=109, FATALITIES=11, SERIOUS_FATAL_RATE=66.88,
REGION_RANK=1),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=448, SERIOUS_FATAL_CRASHES=271,
SERIOUS_INJURIES=295, FATALITIES=31, SERIOUS_FATAL_RATE=60.49,
REGION_RANK=2),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=389, SERIOUS_FATAL_CRASHES=218,
SERIOUS_INJURIES=281, FATALITIES=20, SERIOUS_FATAL_RATE=56.04,
REGION_RANK=3),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=370, SERIOUS_FATAL_CRASHES=190,
SERIOUS_INJURIES=196, FATALITIES=28, SERIOUS_FATAL_RATE=51.35,
REGION_RANK=1),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=113, SERIOUS_FATAL_CRASHES=53,
SERIOUS_INJURIES=51, FATALITIES=11, SERIOUS_FATAL_RATE=46.9,
REGION_RANK=2),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',
TOTAL_CRASHES=930, SERIOUS_FATAL_CRASHES=414, SERIOUS_INJURIES=452,
FATALITIES=48, SERIOUS_FATAL_RATE=44.52, REGION_RANK=3),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=243, SERIOUS_FATAL_CRASHES=135,
SERIOUS_INJURIES=137, FATALITIES=21, SERIOUS_FATAL_RATE=55.56,
REGION_RANK=1),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=126, SERIOUS_FATAL_CRASHES=69,
SERIOUS_INJURIES=76, FATALITIES=11, SERIOUS_FATAL_RATE=54.76,
REGION_RANK=2),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=546, SERIOUS_FATAL_CRASHES=292,
SERIOUS_INJURIES=390, FATALITIES=44, SERIOUS_FATAL_RATE=53.48,
REGION_RANK=3),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=427, SERIOUS_FATAL_CRASHES=225,
SERIOUS_INJURIES=256, FATALITIES=12, SERIOUS_FATAL_RATE=52.69,
REGION_RANK=1),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=288, SERIOUS_FATAL_CRASHES=146,
SERIOUS_INJURIES=149, FATALITIES=14, SERIOUS_FATAL_RATE=50.69,
REGION_RANK=2),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2240, SERIOUS_FATAL_CRASHES=1105,
SERIOUS_INJURIES=1229, FATALITIES=104, SERIOUS_FATAL_RATE=49.33,
REGION_RANK=3),
Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='HIGHWAY',
TOTAL_CRASHES=118, SERIOUS_FATAL_CRASHES=70, SERIOUS_INJURIES=70,
FATALITIES=11, SERIOUS_FATAL_RATE=59.32, REGION_RANK=1),

```

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Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='ROAD',
TOTAL_CRASHES=436, SERIOUS_FATAL_CRASHES=236, SERIOUS_INJURIES=250,
FATALITIES=34, SERIOUS_FATAL_RATE=54.13, REGION_RANK=2),
Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='FREEWAY',
TOTAL_CRASHES=306, SERIOUS_FATAL_CRASHES=159, SERIOUS_INJURIES=208,
FATALITIES=21, SERIOUS_FATAL_RATE=51.96, REGION_RANK=3),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=133, SERIOUS_FATAL_CRASHES=85,
SERIOUS_INJURIES=87, FATALITIES=9, SERIOUS_FATAL_RATE=63.91,
REGION_RANK=1),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2063, SERIOUS_FATAL_CRASHES=1070,
SERIOUS_INJURIES=1208, FATALITIES=55, SERIOUS_FATAL_RATE=51.87,
REGION_RANK=2),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=217, SERIOUS_FATAL_CRASHES=99,
SERIOUS_INJURIES=121, FATALITIES=6, SERIOUS_FATAL_RATE=45.62,
REGION_RANK=3),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=141, SERIOUS_FATAL_CRASHES=74,
SERIOUS_INJURIES=72, FATALITIES=17, SERIOUS_FATAL_RATE=52.48,
REGION_RANK=1),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=329, SERIOUS_FATAL_CRASHES=171,
SERIOUS_INJURIES=181, FATALITIES=21, SERIOUS_FATAL_RATE=51.98,
REGION_RANK=2),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=894, SERIOUS_FATAL_CRASHES=441,
SERIOUS_INJURIES=483, FATALITIES=44, SERIOUS_FATAL_RATE=49.33,
REGION_RANK=3)]

```

```
%%time
```

```

benchmark_df = crashes_selected \
    .withColumn(
        "HOUR",
        substring("ACCIDENT_TIME", 1, 2).cast("int")
    ) \
    .withColumn(
        "TIME_PERIOD",
        when(col("HOUR") < 6, "Overnight")
        .when(col("HOUR") < 12, "Morning")
        .when(col("HOUR") < 18, "Afternoon")
        .otherwise("Evening")
    ) \
    .withColumn(
        "SERIOUS_FATAL_CRASH",
        when(
            (col("SERIOUSINJURY") > 0) | (col("FATALITY") > 0),
            1
        )
    )

```

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        ).otherwise(0)
    ) \
    .groupBy(
        "DTP_REGION",
        "TIME_PERIOD",
        "ROAD_TYPE"
    ) \
    .agg(
        count("ACCIDENT_NO").alias("TOTAL_CRASHES"),
        sum("SERIOUS_FATAL_CRASH").alias("SERIOUS_FATAL_CRASHES"),
        sum("SERIOUS_INJURY").alias("SERIOUS_INJURIES"),
        sum("FATALITY").alias("FATALITIES")
    ) \
    .withColumn(
        "SERIOUS_FATAL_RATE",
        round(
            col("SERIOUS_FATAL_CRASHES") /
            col("TOTAL_CRASHES") * 100,
            2
        )
    ) \
    .filter(col("TOTAL_CRASHES") >= 100)

window_spec = Window \
    .partitionBy("DTP_REGION") \
    .orderBy(col("SERIOUS_FATAL_RATE").desc())

benchmark_df = benchmark_df.withColumn(
    "REGION_RANK",
    row_number().over(window_spec)
)

benchmark_df = benchmark_df \
    .filter(col("REGION_RANK") <= 3) \
    .orderBy("DTP_REGION", "REGION_RANK")

benchmark_df.collect()

```

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CPU times: user 10.6 ms, sys: 6.43 ms, total: 17 ms
Wall time: 375 ms

```

```

[Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=103,
SERIOUS_INJURIES=109, FATALITIES=11, SERIOUS_FATAL_RATE=66.88,
REGION_RANK=1),
 Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=448, SERIOUS_FATAL_CRASHES=271,
SERIOUS_INJURIES=295, FATALITIES=31, SERIOUS_FATAL_RATE=60.49,
REGION_RANK=2),
 Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',

```

ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=389, SERIOUS_FATAL_CRASHES=218,
 SERIOUS_INJURIES=281, FATALITIES=20, SERIOUS_FATAL_RATE=56.04,
 REGION_RANK=3),
 Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
 ROAD_TYPE='ROAD', TOTAL_CRASHES=370, SERIOUS_FATAL_CRASHES=190,
 SERIOUS_INJURIES=196, FATALITIES=28, SERIOUS_FATAL_RATE=51.35,
 REGION_RANK=1),
 Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=113, SERIOUS_FATAL_CRASHES=53,
 SERIOUS_INJURIES=51, FATALITIES=11, SERIOUS_FATAL_RATE=46.9,
 REGION_RANK=2),
 Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',
 TOTAL_CRASHES=930, SERIOUS_FATAL_CRASHES=414, SERIOUS_INJURIES=452,
 FATALITIES=48, SERIOUS_FATAL_RATE=44.52, REGION_RANK=3),
 Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
 ROAD_TYPE='ROAD', TOTAL_CRASHES=243, SERIOUS_FATAL_CRASHES=135,
 SERIOUS_INJURIES=137, FATALITIES=21, SERIOUS_FATAL_RATE=55.56,
 REGION_RANK=1),
 Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=126, SERIOUS_FATAL_CRASHES=69,
 SERIOUS_INJURIES=76, FATALITIES=11, SERIOUS_FATAL_RATE=54.76,
 REGION_RANK=2),
 Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=546, SERIOUS_FATAL_CRASHES=292,
 SERIOUS_INJURIES=390, FATALITIES=44, SERIOUS_FATAL_RATE=53.48,
 REGION_RANK=3),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=427, SERIOUS_FATAL_CRASHES=225,
 SERIOUS_INJURIES=256, FATALITIES=12, SERIOUS_FATAL_RATE=52.69,
 REGION_RANK=1),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='FREEWAY', TOTAL_CRASHES=288, SERIOUS_FATAL_CRASHES=146,
 SERIOUS_INJURIES=149, FATALITIES=14, SERIOUS_FATAL_RATE=50.69,
 REGION_RANK=2),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='ROAD', TOTAL_CRASHES=2240, SERIOUS_FATAL_CRASHES=1105,
 SERIOUS_INJURIES=1229, FATALITIES=104, SERIOUS_FATAL_RATE=49.33,
 REGION_RANK=3),
 Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='HIGHWAY',
 TOTAL_CRASHES=118, SERIOUS_FATAL_CRASHES=70, SERIOUS_INJURIES=70,
 FATALITIES=11, SERIOUS_FATAL_RATE=59.32, REGION_RANK=1),
 Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='ROAD',
 TOTAL_CRASHES=436, SERIOUS_FATAL_CRASHES=236, SERIOUS_INJURIES=250,
 FATALITIES=34, SERIOUS_FATAL_RATE=54.13, REGION_RANK=2),
 Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='FREEWAY',
 TOTAL_CRASHES=306, SERIOUS_FATAL_CRASHES=159, SERIOUS_INJURIES=208,
 FATALITIES=21, SERIOUS_FATAL_RATE=51.96, REGION_RANK=3),
 Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=133, SERIOUS_FATAL_CRASHES=85,

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SERIOUS_INJURIES=87, FATALITIES=9, SERIOUS_FATAL_RATE=63.91,
REGION_RANK=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2063, SERIOUS_FATAL_CRASHES=1070,
SERIOUS_INJURIES=1208, FATALITIES=55, SERIOUS_FATAL_RATE=51.87,
REGION_RANK=2),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=217, SERIOUS_FATAL_CRASHES=99,
SERIOUS_INJURIES=121, FATALITIES=6, SERIOUS_FATAL_RATE=45.62,
REGION_RANK=3),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=141, SERIOUS_FATAL_CRASHES=74,
SERIOUS_INJURIES=72, FATALITIES=17, SERIOUS_FATAL_RATE=52.48,
REGION_RANK=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=329, SERIOUS_FATAL_CRASHES=171,
SERIOUS_INJURIES=181, FATALITIES=21, SERIOUS_FATAL_RATE=51.98,
REGION_RANK=2),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=894, SERIOUS_FATAL_CRASHES=441,
SERIOUS_INJURIES=483, FATALITIES=44, SERIOUS_FATAL_RATE=49.33,
REGION_RANK=3)]

```

```
%%time
```

```

benchmark_df = crashes_selected \
    .withColumn(
        "HOUR",
        substring("ACCIDENT_TIME", 1, 2).cast("int")
    ) \
    .withColumn(
        "TIME_PERIOD",
        when(col("HOUR") < 6, "Overnight")
        .when(col("HOUR") < 12, "Morning")
        .when(col("HOUR") < 18, "Afternoon")
        .otherwise("Evening")
    ) \
    .withColumn(
        "SERIOUS_FATAL_CRASH",
        when(
            (col("SERIOUSINJURY") > 0) | (col("FATALITY") > 0),
            1
        ).otherwise(0)
    ) \
    .groupBy(
        "DTP_REGION",
        "TIME_PERIOD",
        "ROAD_TYPE"
    ) \
    .agg(

```

```

        count("ACCIDENT_NO").alias("TOTAL_CRASHES"),
        sum("SERIOUS_FATAL_CRASH").alias("SERIOUS_FATAL_CRASHES"),
        sum("SERIOUS_INJURY").alias("SERIOUS_INJURIES"),
        sum("FATALITY").alias("FATALITIES")
    ) \
    .withColumn(
        "SERIOUS_FATAL_RATE",
        round(
            col("SERIOUS_FATAL_CRASHES") /
            col("TOTAL_CRASHES") * 100,
            2
        )
    ) \
    .filter(col("TOTAL_CRASHES") >= 100)

window_spec = Window \
    .partitionBy("DTP_REGION") \
    .orderBy(col("SERIOUS_FATAL_RATE").desc())

benchmark_df = benchmark_df.withColumn(
    "REGION_RANK",
    row_number().over(window_spec)
)

benchmark_df = benchmark_df \
    .filter(col("REGION_RANK") <= 3) \
    .orderBy("DTP_REGION", "REGION_RANK")

benchmark_df.collect()

CPU times: user 9.64 ms, sys: 5.76 ms, total: 15.4 ms
Wall time: 330 ms

[Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=103,
SERIOUS_INJURIES=109, FATALITIES=11, SERIOUS_FATAL_RATE=66.88,
REGION_RANK=1),
 Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=448, SERIOUS_FATAL_CRASHES=271,
SERIOUS_INJURIES=295, FATALITIES=31, SERIOUS_FATAL_RATE=60.49,
REGION_RANK=2),
 Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=389, SERIOUS_FATAL_CRASHES=218,
SERIOUS_INJURIES=281, FATALITIES=20, SERIOUS_FATAL_RATE=56.04,
REGION_RANK=3),
 Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=370, SERIOUS_FATAL_CRASHES=190,
SERIOUS_INJURIES=196, FATALITIES=28, SERIOUS_FATAL_RATE=51.35,
REGION_RANK=1),
 Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
```

ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=113, SERIOUS_FATAL_CRASHES=53,
 SERIOUS_INJURIES=51, FATALITIES=11, SERIOUS_FATAL_RATE=46.9,
 REGION_RANK=2),
 Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',
 TOTAL_CRASHES=930, SERIOUS_FATAL_CRASHES=414, SERIOUS_INJURIES=452,
 FATALITIES=48, SERIOUS_FATAL_RATE=44.52, REGION_RANK=3),
 Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
 ROAD_TYPE='ROAD', TOTAL_CRASHES=243, SERIOUS_FATAL_CRASHES=135,
 SERIOUS_INJURIES=137, FATALITIES=21, SERIOUS_FATAL_RATE=55.56,
 REGION_RANK=1),
 Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=126, SERIOUS_FATAL_CRASHES=69,
 SERIOUS_INJURIES=76, FATALITIES=11, SERIOUS_FATAL_RATE=54.76,
 REGION_RANK=2),
 Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=546, SERIOUS_FATAL_CRASHES=292,
 SERIOUS_INJURIES=390, FATALITIES=44, SERIOUS_FATAL_RATE=53.48,
 REGION_RANK=3),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=427, SERIOUS_FATAL_CRASHES=225,
 SERIOUS_INJURIES=256, FATALITIES=12, SERIOUS_FATAL_RATE=52.69,
 REGION_RANK=1),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='FREEWAY', TOTAL_CRASHES=288, SERIOUS_FATAL_CRASHES=146,
 SERIOUS_INJURIES=149, FATALITIES=14, SERIOUS_FATAL_RATE=50.69,
 REGION_RANK=2),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='ROAD', TOTAL_CRASHES=2240, SERIOUS_FATAL_CRASHES=1105,
 SERIOUS_INJURIES=1229, FATALITIES=104, SERIOUS_FATAL_RATE=49.33,
 REGION_RANK=3),
 Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='HIGHWAY',
 TOTAL_CRASHES=118, SERIOUS_FATAL_CRASHES=70, SERIOUS_INJURIES=70,
 FATALITIES=11, SERIOUS_FATAL_RATE=59.32, REGION_RANK=1),
 Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='ROAD',
 TOTAL_CRASHES=436, SERIOUS_FATAL_CRASHES=236, SERIOUS_INJURIES=250,
 FATALITIES=34, SERIOUS_FATAL_RATE=54.13, REGION_RANK=2),
 Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='FREEWAY',
 TOTAL_CRASHES=306, SERIOUS_FATAL_CRASHES=159, SERIOUS_INJURIES=208,
 FATALITIES=21, SERIOUS_FATAL_RATE=51.96, REGION_RANK=3),
 Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=133, SERIOUS_FATAL_CRASHES=85,
 SERIOUS_INJURIES=87, FATALITIES=9, SERIOUS_FATAL_RATE=63.91,
 REGION_RANK=1),
 Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='ROAD', TOTAL_CRASHES=2063, SERIOUS_FATAL_CRASHES=1070,
 SERIOUS_INJURIES=1208, FATALITIES=55, SERIOUS_FATAL_RATE=51.87,
 REGION_RANK=2),
 Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='FREEWAY', TOTAL_CRASHES=217, SERIOUS_FATAL_CRASHES=99,

```
SERIOUS_INJURIES=121, FATALITIES=6, SERIOUS_FATAL_RATE=45.62,
REGION_RANK=3),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=141, SERIOUS_FATAL_CRASHES=74,
SERIOUS_INJURIES=72, FATALITIES=17, SERIOUS_FATAL_RATE=52.48,
REGION_RANK=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=329, SERIOUS_FATAL_CRASHES=171,
SERIOUS_INJURIES=181, FATALITIES=21, SERIOUS_FATAL_RATE=51.98,
REGION_RANK=2),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=894, SERIOUS_FATAL_CRASHES=441,
SERIOUS_INJURIES=483, FATALITIES=44, SERIOUS_FATAL_RATE=49.33,
REGION_RANK=3)]
```

SQL Benchmarking

```
%%time
```

```
sql_benchmark = spark.sql(sql_query)
```

```
sql_benchmark.collect()
```

```
CPU times: user 1.51 ms, sys: 1.19 ms, total: 2.7 ms
```

```
Wall time: 445 ms
```

```
[Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=103,
SERIOUS_INJURIES=109, FATALITIES=11, SERIOUS_FATAL_RATE=66.88,
REGION_RANK=1),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=448, SERIOUS_FATAL_CRASHES=271,
SERIOUS_INJURIES=295, FATALITIES=31, SERIOUS_FATAL_RATE=60.49,
REGION_RANK=2),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=389, SERIOUS_FATAL_CRASHES=218,
SERIOUS_INJURIES=281, FATALITIES=20, SERIOUS_FATAL_RATE=56.04,
REGION_RANK=3),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=370, SERIOUS_FATAL_CRASHES=190,
SERIOUS_INJURIES=196, FATALITIES=28, SERIOUS_FATAL_RATE=51.35,
REGION_RANK=1),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=113, SERIOUS_FATAL_CRASHES=53,
SERIOUS_INJURIES=51, FATALITIES=11, SERIOUS_FATAL_RATE=46.9,
REGION_RANK=2),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',
TOTAL_CRASHES=930, SERIOUS_FATAL_CRASHES=414, SERIOUS_INJURIES=452,
FATALITIES=48, SERIOUS_FATAL_RATE=44.52, REGION_RANK=3),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
```

ROAD_TYPE='ROAD', TOTAL_CRASHES=243, SERIOUS_FATAL_CRASHES=135,
 SERIOUS_INJURIES=137, FATALITIES=21, SERIOUS_FATAL_RATE=55.56,
 REGION_RANK=1),
 Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=126, SERIOUS_FATAL_CRASHES=69,
 SERIOUS_INJURIES=76, FATALITIES=11, SERIOUS_FATAL_RATE=54.76,
 REGION_RANK=2),
 Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=546, SERIOUS_FATAL_CRASHES=292,
 SERIOUS_INJURIES=390, FATALITIES=44, SERIOUS_FATAL_RATE=53.48,
 REGION_RANK=3),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=427, SERIOUS_FATAL_CRASHES=225,
 SERIOUS_INJURIES=256, FATALITIES=12, SERIOUS_FATAL_RATE=52.69,
 REGION_RANK=1),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='FREEWAY', TOTAL_CRASHES=288, SERIOUS_FATAL_CRASHES=146,
 SERIOUS_INJURIES=149, FATALITIES=14, SERIOUS_FATAL_RATE=50.69,
 REGION_RANK=2),
 Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='ROAD', TOTAL_CRASHES=2240, SERIOUS_FATAL_CRASHES=1105,
 SERIOUS_INJURIES=1229, FATALITIES=104, SERIOUS_FATAL_RATE=49.33,
 REGION_RANK=3),
 Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='HIGHWAY',
 TOTAL_CRASHES=118, SERIOUS_FATAL_CRASHES=70, SERIOUS_INJURIES=70,
 FATALITIES=11, SERIOUS_FATAL_RATE=59.32, REGION_RANK=1),
 Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='ROAD',
 TOTAL_CRASHES=436, SERIOUS_FATAL_CRASHES=236, SERIOUS_INJURIES=250,
 FATALITIES=34, SERIOUS_FATAL_RATE=54.13, REGION_RANK=2),
 Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='FREEWAY',
 TOTAL_CRASHES=306, SERIOUS_FATAL_CRASHES=159, SERIOUS_INJURIES=208,
 FATALITIES=21, SERIOUS_FATAL_RATE=51.96, REGION_RANK=3),
 Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=133, SERIOUS_FATAL_CRASHES=85,
 SERIOUS_INJURIES=87, FATALITIES=9, SERIOUS_FATAL_RATE=63.91,
 REGION_RANK=1),
 Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='ROAD', TOTAL_CRASHES=2063, SERIOUS_FATAL_CRASHES=1070,
 SERIOUS_INJURIES=1208, FATALITIES=55, SERIOUS_FATAL_RATE=51.87,
 REGION_RANK=2),
 Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
 ROAD_TYPE='FREEWAY', TOTAL_CRASHES=217, SERIOUS_FATAL_CRASHES=99,
 SERIOUS_INJURIES=121, FATALITIES=6, SERIOUS_FATAL_RATE=45.62,
 REGION_RANK=3),
 Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
 ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=141, SERIOUS_FATAL_CRASHES=74,
 SERIOUS_INJURIES=72, FATALITIES=17, SERIOUS_FATAL_RATE=52.48,
 REGION_RANK=1),
 Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',

```
ROAD_TYPE='ROAD', TOTAL_CRASHES=329, SERIOUS_FATAL_CRASHES=171,  
SERIOUS_INJURIES=181, FATALITIES=21, SERIOUS_FATAL_RATE=51.98,  
REGION_RANK=2),  
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',  
ROAD_TYPE='ROAD', TOTAL_CRASHES=894, SERIOUS_FATAL_CRASHES=441,  
SERIOUS_INJURIES=483, FATALITIES=44, SERIOUS_FATAL_RATE=49.33,  
REGION_RANK=3)]
```

```
%%time
```

```
sql_benchmark = spark.sql(sql_query)
```

```
# Force complete execution  
sql_benchmark.collect()
```

```
CPU times: user 1.76 ms, sys: 767 µs, total: 2.53 ms  
Wall time: 342 ms
```

```
[Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=103,  
SERIOUS_INJURIES=109, FATALITIES=11, SERIOUS_FATAL_RATE=66.88,  
REGION_RANK=1),  
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',  
ROAD_TYPE='ROAD', TOTAL_CRASHES=448, SERIOUS_FATAL_CRASHES=271,  
SERIOUS_INJURIES=295, FATALITIES=31, SERIOUS_FATAL_RATE=60.49,  
REGION_RANK=2),  
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=389, SERIOUS_FATAL_CRASHES=218,  
SERIOUS_INJURIES=281, FATALITIES=20, SERIOUS_FATAL_RATE=56.04,  
REGION_RANK=3),  
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',  
ROAD_TYPE='ROAD', TOTAL_CRASHES=370, SERIOUS_FATAL_CRASHES=190,  
SERIOUS_INJURIES=196, FATALITIES=28, SERIOUS_FATAL_RATE=51.35,  
REGION_RANK=1),  
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=113, SERIOUS_FATAL_CRASHES=53,  
SERIOUS_INJURIES=51, FATALITIES=11, SERIOUS_FATAL_RATE=46.9,  
REGION_RANK=2),  
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',  
TOTAL_CRASHES=930, SERIOUS_FATAL_CRASHES=414, SERIOUS_INJURIES=452,  
FATALITIES=48, SERIOUS_FATAL_RATE=44.52, REGION_RANK=3),  
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',  
ROAD_TYPE='ROAD', TOTAL_CRASHES=243, SERIOUS_FATAL_CRASHES=135,  
SERIOUS_INJURIES=137, FATALITIES=21, SERIOUS_FATAL_RATE=55.56,  
REGION_RANK=1),  
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=126, SERIOUS_FATAL_CRASHES=69,  
SERIOUS_INJURIES=76, FATALITIES=11, SERIOUS_FATAL_RATE=54.76,  
REGION_RANK=2),  
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
```

```

ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=546, SERIOUS_FATAL_CRASHES=292,
SERIOUS_INJURIES=390, FATALITIES=44, SERIOUS_FATAL_RATE=53.48,
REGION_RANK=3),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=427, SERIOUS_FATAL_CRASHES=225,
SERIOUS_INJURIES=256, FATALITIES=12, SERIOUS_FATAL_RATE=52.69,
REGION_RANK=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=288, SERIOUS_FATAL_CRASHES=146,
SERIOUS_INJURIES=149, FATALITIES=14, SERIOUS_FATAL_RATE=50.69,
REGION_RANK=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2240, SERIOUS_FATAL_CRASHES=1105,
SERIOUS_INJURIES=1229, FATALITIES=104, SERIOUS_FATAL_RATE=49.33,
REGION_RANK=3),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='HIGHWAY',
TOTAL_CRASHES=118, SERIOUS_FATAL_CRASHES=70, SERIOUS_INJURIES=70,
FATALITIES=11, SERIOUS_FATAL_RATE=59.32, REGION_RANK=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='ROAD',
TOTAL_CRASHES=436, SERIOUS_FATAL_CRASHES=236, SERIOUS_INJURIES=250,
FATALITIES=34, SERIOUS_FATAL_RATE=54.13, REGION_RANK=2),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='FREEWAY',
TOTAL_CRASHES=306, SERIOUS_FATAL_CRASHES=159, SERIOUS_INJURIES=208,
FATALITIES=21, SERIOUS_FATAL_RATE=51.96, REGION_RANK=3),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=133, SERIOUS_FATAL_CRASHES=85,
SERIOUS_INJURIES=87, FATALITIES=9, SERIOUS_FATAL_RATE=63.91,
REGION_RANK=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2063, SERIOUS_FATAL_CRASHES=1070,
SERIOUS_INJURIES=1208, FATALITIES=55, SERIOUS_FATAL_RATE=51.87,
REGION_RANK=2),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=217, SERIOUS_FATAL_CRASHES=99,
SERIOUS_INJURIES=121, FATALITIES=6, SERIOUS_FATAL_RATE=45.62,
REGION_RANK=3),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=141, SERIOUS_FATAL_CRASHES=74,
SERIOUS_INJURIES=72, FATALITIES=17, SERIOUS_FATAL_RATE=52.48,
REGION_RANK=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=329, SERIOUS_FATAL_CRASHES=171,
SERIOUS_INJURIES=181, FATALITIES=21, SERIOUS_FATAL_RATE=51.98,
REGION_RANK=2),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=894, SERIOUS_FATAL_CRASHES=441,
SERIOUS_INJURIES=483, FATALITIES=44, SERIOUS_FATAL_RATE=49.33,
REGION_RANK=3)]

```

```
%%time
```

```
sql_benchmark = spark.sql(sql_query)
```

```
# Force complete execution
```

```
sql_benchmark.collect()
```

```
CPU times: user 1.58 ms, sys: 822 µs, total: 2.41 ms
```

```
Wall time: 346 ms
```

```
[Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=103,  
SERIOUS_INJURIES=109, FATALITIES=11, SERIOUS_FATAL_RATE=66.88,  
REGION_RANK=1),  
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',  
ROAD_TYPE='ROAD', TOTAL_CRASHES=448, SERIOUS_FATAL_CRASHES=271,  
SERIOUS_INJURIES=295, FATALITIES=31, SERIOUS_FATAL_RATE=60.49,  
REGION_RANK=2),  
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=389, SERIOUS_FATAL_CRASHES=218,  
SERIOUS_INJURIES=281, FATALITIES=20, SERIOUS_FATAL_RATE=56.04,  
REGION_RANK=3),  
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',  
ROAD_TYPE='ROAD', TOTAL_CRASHES=370, SERIOUS_FATAL_CRASHES=190,  
SERIOUS_INJURIES=196, FATALITIES=28, SERIOUS_FATAL_RATE=51.35,  
REGION_RANK=1),  
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=113, SERIOUS_FATAL_CRASHES=53,  
SERIOUS_INJURIES=51, FATALITIES=11, SERIOUS_FATAL_RATE=46.9,  
REGION_RANK=2),  
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',  
TOTAL_CRASHES=930, SERIOUS_FATAL_CRASHES=414, SERIOUS_INJURIES=452,  
FATALITIES=48, SERIOUS_FATAL_RATE=44.52, REGION_RANK=3),  
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',  
ROAD_TYPE='ROAD', TOTAL_CRASHES=243, SERIOUS_FATAL_CRASHES=135,  
SERIOUS_INJURIES=137, FATALITIES=21, SERIOUS_FATAL_RATE=55.56,  
REGION_RANK=1),  
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=126, SERIOUS_FATAL_CRASHES=69,  
SERIOUS_INJURIES=76, FATALITIES=11, SERIOUS_FATAL_RATE=54.76,  
REGION_RANK=2),  
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=546, SERIOUS_FATAL_CRASHES=292,  
SERIOUS_INJURIES=390, FATALITIES=44, SERIOUS_FATAL_RATE=53.48,  
REGION_RANK=3),  
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=427, SERIOUS_FATAL_CRASHES=225,  
SERIOUS_INJURIES=256, FATALITIES=12, SERIOUS_FATAL_RATE=52.69,  
REGION_RANK=1),  
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
```

```

ROAD_TYPE='FREEWAY', TOTAL_CRASHES=288, SERIOUS_FATAL_CRASHES=146,
SERIOUS_INJURIES=149, FATALITIES=14, SERIOUS_FATAL_RATE=50.69,
REGION_RANK=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2240, SERIOUS_FATAL_CRASHES=1105,
SERIOUS_INJURIES=1229, FATALITIES=104, SERIOUS_FATAL_RATE=49.33,
REGION_RANK=3),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='HIGHWAY',
TOTAL_CRASHES=118, SERIOUS_FATAL_CRASHES=70, SERIOUS_INJURIES=70,
FATALITIES=11, SERIOUS_FATAL_RATE=59.32, REGION_RANK=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='ROAD',
TOTAL_CRASHES=436, SERIOUS_FATAL_CRASHES=236, SERIOUS_INJURIES=250,
FATALITIES=34, SERIOUS_FATAL_RATE=54.13, REGION_RANK=2),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='FREEWAY',
TOTAL_CRASHES=306, SERIOUS_FATAL_CRASHES=159, SERIOUS_INJURIES=208,
FATALITIES=21, SERIOUS_FATAL_RATE=51.96, REGION_RANK=3),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=133, SERIOUS_FATAL_CRASHES=85,
SERIOUS_INJURIES=87, FATALITIES=9, SERIOUS_FATAL_RATE=63.91,
REGION_RANK=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2063, SERIOUS_FATAL_CRASHES=1070,
SERIOUS_INJURIES=1208, FATALITIES=55, SERIOUS_FATAL_RATE=51.87,
REGION_RANK=2),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=217, SERIOUS_FATAL_CRASHES=99,
SERIOUS_INJURIES=121, FATALITIES=6, SERIOUS_FATAL_RATE=45.62,
REGION_RANK=3),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=141, SERIOUS_FATAL_CRASHES=74,
SERIOUS_INJURIES=72, FATALITIES=17, SERIOUS_FATAL_RATE=52.48,
REGION_RANK=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=329, SERIOUS_FATAL_CRASHES=171,
SERIOUS_INJURIES=181, FATALITIES=21, SERIOUS_FATAL_RATE=51.98,
REGION_RANK=2),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=894, SERIOUS_FATAL_CRASHES=441,
SERIOUS_INJURIES=483, FATALITIES=44, SERIOUS_FATAL_RATE=49.33,
REGION_RANK=3)]

```

Summary Table

```

import statistics

df_times = [324, 375, 330]
sql_times = [445, 342, 346]

```

```
print("DataFrame median (ms):", statistics.median(df_times))
print("Spark SQL median (ms):", statistics.median(sql_times))
```

```
DataFrame median: 330
Spark SQL median: 346
```

Discussion

Benchmarking was done using Apache Spark 4.2.0 locally on a single machine, so was done using local CPU resources. Spark default memory configuration was used provide by the environment. Since the experiments were run locally, the results may be different compared to if a distributed cluster was used to execute the query.

The execution times show that the Dataframe API was slightly faster than the SQL implementation in the local environment. The median Dataframe runtime was 330ms while the SQL median is 346ms, which is a difference of 16ms, or the Dataframe implementation is 4.6% faster than that of the SQL implementation. The first SQL runtime was substantially slower at 445ms which could be due to JVM warm-up, so it was important to execute the implementation more than once and take the median.

Both the SQL and Dataframe implementations are processed through the Spark Catalyst optimiser, but there can be differences in performance that are due to query construction, optimisation, execution planning and runtime overhead. Both implementations were designed to perform similar transformations, including aggregation, filtering, window operation, and sorting, so they should generate comparable workloads. The difference between the median execution time shows that neither implementation has a big performance advantage over the other. Also, since the execution was done in a local environment, it cannot be representative of a large scale performance. Shuffle volume, partitioning, serialisation and resources have a bigger impact when executed on a cluster.

3. Execution plan interpretation

```
aggregated.explain(extended=True)
```

```
== Parsed Logical Plan ==
'Aggregate ['DTP_REGION', 'TIME_PERIOD', 'ROAD_TYPE'], ['DTP_REGION',
'TIME_PERIOD', 'ROAD_TYPE', 'count(distinct 'ACCIDENT_NO') AS
TOTAL_CRASHES#397, 'sum('SERIOUS_FATAL_CRASH) AS
SERIOUS_FATAL_CRASHES#398, 'sum('SERIOUSINJURY) AS
SERIOUS_INJURIES#399, 'sum('FATALITY) AS FATALITIES#400]
+- Project [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78, TIME_PERIOD#79,
CASE WHEN ((SERIOUSINJURY#25 > 0) OR (FATALITY#24 > 0)) THEN 1 ELSE 0
END AS SERIOUS_FATAL_CRASH#91]
  +- Project [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78, CASE WHEN
(HOUR#78 < 6) THEN Overnight WHEN (HOUR#78 < 12) THEN Morning WHEN
(HOUR#78 < 18) THEN Afternoon ELSE Evening END AS TIME_PERIOD#79]
    +- Project [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24,
```

```

cast(substring(ACCIDENT_TIME#2, 1, 2) as int) AS HOUR#78]
+- Project [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24]
+- Relation
[ACCIDENT_NO#0,ACCIDENT_DATE#1,ACCIDENT_TIME#2,ACCIDENT_TYPE#3,DAY_OF_
WEEK#4,DCA_CODE#5,DCA_CODE_DESCRIPTION#6,LIGHT_CONDITION#7,POLICE_ATTEN
D#8,ROAD_GEOMETRY#9,SEVERITY#10,SPEED_ZONE#11,RUN_OFFROAD#12,ROAD_NAME#13,ROAD_TYPE#14,ROAD_ROUTE_1#15,LGA_NAME#16,DTP_REGION#17,LATITUDE#18,
LONGITUDE#19,VICGRID_X#20,VICGRID_Y#21,TOTAL_PERSONS#22,INJ_OR_FATALITY#23,FATALITY#24,... 27 more fields] csv

```

== Analyzed Logical Plan ==

```

DTP_REGION: string, TIME_PERIOD: string, ROAD_TYPE: string,
TOTAL_CRASHES: bigint, SERIOUS_FATAL_CRASHES: bigint,
SERIOUS_INJURIES: bigint, FATALITIES: bigint
Aggregate [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14],
[DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, count(distinct
ACCIDENT_NO#0) AS TOTAL_CRASHES#397L, sum(SERIOUS_FATAL_CRASH#91) AS
SERIOUS_FATAL_CRASHES#398L, sum(SERIOUSINJURY#25) AS
SERIOUS_INJURIES#399L, sum(FATALITY#24) AS FATALITIES#400L]
+- Project [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78, TIME_PERIOD#79,
CASE WHEN ((SERIOUSINJURY#25 > 0) OR (FATALITY#24 > 0)) THEN 1 ELSE 0
END AS SERIOUS_FATAL_CRASH#91]

```

```

+- Project [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78, CASE WHEN
(HOUR#78 < 6) THEN Overnight WHEN (HOUR#78 < 12) THEN Morning WHEN
(HOUR#78 < 18) THEN Afternoon ELSE Evening END AS TIME_PERIOD#79]

```

```

+- Project [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24,
cast(substring(ACCIDENT_TIME#2, 1, 2) as int) AS HOUR#78]
+- Project [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24]
+- Relation
[ACCIDENT_NO#0,ACCIDENT_DATE#1,ACCIDENT_TIME#2,ACCIDENT_TYPE#3,DAY_OF_
WEEK#4,DCA_CODE#5,DCA_CODE_DESCRIPTION#6,LIGHT_CONDITION#7,POLICE_ATTEN
D#8,ROAD_GEOMETRY#9,SEVERITY#10,SPEED_ZONE#11,RUN_OFFROAD#12,ROAD_NAME#13,ROAD_TYPE#14,ROAD_ROUTE_1#15,LGA_NAME#16,DTP_REGION#17,LATITUDE#18,
LONGITUDE#19,VICGRID_X#20,VICGRID_Y#21,TOTAL_PERSONS#22,INJ_OR_FATALITY#23,FATALITY#24,... 27 more fields] csv

```

== Optimized Logical Plan ==

```

Aggregate [DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14],
[DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, count(distinct
ACCIDENT_NO#0) AS TOTAL_CRASHES#397L, sum(SERIOUS_FATAL_CRASH#91) AS
SERIOUS_FATAL_CRASHES#398L, sum(SERIOUSINJURY#25) AS
SERIOUS_INJURIES#399L, sum(FATALITY#24) AS FATALITIES#400L]
+- Project [ACCIDENT_NO#0, DTP_REGION#17, ROAD_TYPE#14,
SERIOUSINJURY#25, FATALITY#24, TIME_PERIOD#79, SERIOUS_FATAL_CRASH#91]

```

```

+- InMemoryRelation [ACCIDENT_NO#0, ACCIDENT_TIME#2, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78, TIME_PERIOD#79,
SERIOUS_FATAL_CRASH#91], StorageLevel(disk, memory, deserialized, 1
replicas)
+- *(1) Project [ACCIDENT_NO#0, ACCIDENT_TIME#2,
DTP_REGION#17, ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, HOUR#78,
CASE WHEN (HOUR#78 < 6) THEN Overnight WHEN (HOUR#78 < 12) THEN
Morning WHEN (HOUR#78 < 18) THEN Afternoon ELSE Evening END AS
TIME_PERIOD#79, CASE WHEN ((SERIOUSINJURY#25 > 0) OR (FATALITY#24 >
0)) THEN 1 ELSE 0 END AS SERIOUS_FATAL_CRASH#91]
+- *(1) Project [ACCIDENT_NO#0, ACCIDENT_TIME#2,
DTP_REGION#17, ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24,
cast(substring(ACCIDENT_TIME#2, 1, 2) as int) AS HOUR#78]
+- FileScan csv
[ACCIDENT_NO#0,ACCIDENT_TIME#2,ROAD_TYPE#14,DTP_REGION#17,FATALITY#24,
SERIOUSINJURY#25] Batched: false, DataFilters: [], Format: CSV,
Location: InMemoryFileIndex(1
paths)[file:/Users/stefangarevski/Desktop/Data Processing for Big Data
IT0520..., PartitionFilters: [], PushedFilters: [], ReadSchema:
struct<ACCIDENT_NO:string,ACCIDENT_TIME:string,ROAD_TYPE:string,DTP_RE
GION:string,FATALITY:int,SE...

== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- HashAggregate(keys=[DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14],
functions=[sum(SERIOUS_FATAL_CRASH#91), sum(SERIOUSINJURY#25),
sum(FATALITY#24), count(distinct ACCIDENT_NO#0)],
output=[DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14,
TOTAL_CRASHES#397L, SERIOUS_FATAL_CRASHES#398L, SERIOUS_INJURIES#399L,
FATALITIES#400L])
+- Exchange hashpartitioning(DTP_REGION#17, TIME_PERIOD#79,
ROAD_TYPE#14, 200), ENSURE_REQUIREMENTS, [plan_id=2795]
+- HashAggregate(keys=[DTP_REGION#17, TIME_PERIOD#79,
ROAD_TYPE#14], functions=[merge_sum(SERIOUS_FATAL_CRASH#91),
merge_sum(SERIOUSINJURY#25), merge_sum(FATALITY#24),
partial_count(distinct ACCIDENT_NO#0)], output=[DTP_REGION#17,
TIME_PERIOD#79, ROAD_TYPE#14, sum#557L, sum#559L, sum#561L,
count#564L])
+- HashAggregate(keys=[DTP_REGION#17, TIME_PERIOD#79,
ROAD_TYPE#14, ACCIDENT_NO#0],
functions=[merge_sum(SERIOUS_FATAL_CRASH#91),
merge_sum(SERIOUSINJURY#25), merge_sum(FATALITY#24)],
output=[DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0,
sum#557L, sum#559L, sum#561L])
+- Exchange hashpartitioning(DTP_REGION#17,
TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0, 200),
ENSURE_REQUIREMENTS, [plan_id=2791]
+- HashAggregate(keys=[DTP_REGION#17, TIME_PERIOD#79,
ROAD_TYPE#14, ACCIDENT_NO#0],

```

```

functions=[partial_sum(SERIOUS_FATAL_CRASH#91),
partial_sum(SERIOUSINJURY#25), partial_sum(FATALITY#24)],
output=[DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0,
sum#557L, sum#559L, sum#561L])
      +- InMemoryTableScan [ACCIDENT_NO#0, DTP_REGION#17,
ROAD_TYPE#14, SERIOUSINJURY#25, FATALITY#24, TIME_PERIOD#79,
SERIOUS_FATAL_CRASH#91]
      +- InMemoryRelation [ACCIDENT_NO#0,
ACCIDENT_TIME#2, DTP_REGION#17, ROAD_TYPE#14, SERIOUSINJURY#25,
FATALITY#24, HOUR#78, TIME_PERIOD#79, SERIOUS_FATAL_CRASH#91],
StorageLevel(disk, memory, deserialized, 1 replicas)
      +- *(1) Project [ACCIDENT_NO#0,
ACCIDENT_TIME#2, DTP_REGION#17, ROAD_TYPE#14, SERIOUSINJURY#25,
FATALITY#24, HOUR#78, CASE WHEN (HOUR#78 < 6) THEN Overnight WHEN
(HOUR#78 < 12) THEN Morning WHEN (HOUR#78 < 18) THEN Afternoon ELSE
Evening END AS TIME_PERIOD#79, CASE WHEN ((SERIOUSINJURY#25 > 0) OR
(FATALITY#24 > 0)) THEN 1 ELSE 0 END AS SERIOUS_FATAL_CRASH#91]
      +- *(1) Project [ACCIDENT_NO#0,
ACCIDENT_TIME#2, DTP_REGION#17, ROAD_TYPE#14, SERIOUSINJURY#25,
FATALITY#24, cast(substring(ACCIDENT_TIME#2, 1, 2) as int) AS HOUR#78]
      +- FileScan csv
[ACCIDENT_NO#0,ACCIDENT_TIME#2,ROAD_TYPE#14,DTP_REGION#17,FATALITY#24,
SERIOUSINJURY#25] Batched: false, DataFilters: [], Format: CSV,
Location: InMemoryFileIndex(1
paths)[file:/Users/stefangarevski/Desktop/Data Processing for Big Data
IT0520..., PartitionFilters: [], PushedFilters: [], ReadSchema:
struct<ACCIDENT_NO:string,ACCIDENT_TIME:string,ROAD_TYPE:string,DTP_RE
GION:string,FATALITY:int,SE...

```

Physical Plan Annotations

The physical plan contains two exchange hashpartitioning operations and they both represent shuffle stages.

Shuffle 1 happens with the line: Exchange hashpartitioning(DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, ACCIDENT_NO#0, 200). Since there is a HashAggregate right before the shuffle, this shuffle redistributes records according to the grouping columns and ACCIDENT_NO, which is required to combine records that belong to the same distinct accident and grouping combination.

Shuffle 2 happens with the line: Exchange hashpartitioning(DTP_REGION#17, TIME_PERIOD#79, ROAD_TYPE#14, 200). The preceding operation is again the HashAggregate, and this shuffle will redistribute the aggregated records so that all that belong to the same DTP_REGION, TIME_PERIOD, ROAD_TYPE group are processed together by the final aggregation.

Analysis

The execution plan shows the aggregation requires shuffling operation to correctly combine records across the partitions. The first Exchange hashpartitioning operator occurs after a HashAggregate where the data is partitioned using DTP_REGION, TIME_PERIOD, ROAD_TYPE, and ACCOUNT_NO and it is associated with the count(distinct ACCIDENT_NO) calculation. The second shuffle partitions the partially aggregated data using DTP_REGION, TIME_PERIOD, and ROAD_TYPE so the final aggregation can combine all records belonging to each group.

The shuffling is necessary because records that are required for the aggregation could be located in various partitions. Spark will redistribute data across partitions based on the fields that the aggregation is occurring on. This involves network I/O in a distributed cluster as data will probably need to move between executors. In a local execution environment, the shuffling doesn't involve network communication between separate machines, but in a distributed cluster shuffle data may be transferred between executors over the network. This could create task scheduling overhead as Spark must complete the shuffle stage before the aggregation can be completed. The preceding partial HashAggregate reduces the amount of data that needs to be shuffled though.

Completely removing the shuffling might be difficult as the operation uses multiple grouping variables and a distinct count for this analysis. The current plan portrays a necessary cost of performing the aggregation.

4. Spark Web UI (DAG visualisation)

```
aggregated.collect()
```

```
[Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',  
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=55, SERIOUS_FATAL_CRASHES=24,  
SERIOUS_INJURIES=25, FATALITIES=1),  
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',  
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=1519, SERIOUS_FATAL_CRASHES=598,  
SERIOUS_INJURIES=736, FATALITIES=28),  
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',  
ROAD_TYPE='LANE', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=4,  
SERIOUS_INJURIES=4, FATALITIES=0),  
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',  
ROAD_TYPE='RAMP', TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=6,  
SERIOUS_INJURIES=8, FATALITIES=0),  
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',  
ROAD_TYPE='PLACE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,  
SERIOUS_INJURIES=1, FATALITIES=0),  
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',  
ROAD_TYPE=None, TOTAL_CRASHES=539, SERIOUS_FATAL_CRASHES=183,  
SERIOUS_INJURIES=208, FATALITIES=3),  
Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE=None,  
TOTAL_CRASHES=18, SERIOUS_FATAL_CRASHES=8, SERIOUS_INJURIES=15,  
FATALITIES=0),  
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',  
TOTAL_CRASHES=765, SERIOUS_FATAL_CRASHES=407, SERIOUS_INJURIES=469,
```

```

FATALITIES=35),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='GROVE', TOTAL_CRASHES=52, SERIOUS_FATAL_CRASHES=23,
SERIOUS_INJURIES=25, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='PROMENADE', TOTAL_CRASHES=18, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='LANE',
TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=4, SERIOUS_INJURIES=3,
FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='RAMP',
TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=5, SERIOUS_INJURIES=5,
FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='HIGHWAY',
TOTAL_CRASHES=699, SERIOUS_FATAL_CRASHES=356, SERIOUS_INJURIES=452,
FATALITIES=49),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='ALLEY', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=104, SERIOUS_FATAL_CRASHES=46,
SERIOUS_INJURIES=48, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=17, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening',
ROAD_TYPE='STREET', TOTAL_CRASHES=341, SERIOUS_FATAL_CRASHES=98,
SERIOUS_INJURIES=104, FATALITIES=4),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='RAMP',
TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=5, SERIOUS_INJURIES=6,
FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=448, SERIOUS_FATAL_CRASHES=271,
SERIOUS_INJURIES=295, FATALITIES=31),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=127, SERIOUS_FATAL_CRASHES=43,
SERIOUS_INJURIES=44, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='DRIVE',
TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=6, SERIOUS_INJURIES=9,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=1058, SERIOUS_FATAL_CRASHES=425,
SERIOUS_INJURIES=475, FATALITIES=15),

```

```

Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='PLACE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION=None, TIME_PERIOD='Evening', ROAD_TYPE=None,
TOTAL_CRASHES=19, SERIOUS_FATAL_CRASHES=7, SERIOUS_INJURIES=7,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=6, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='RISE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=1135, SERIOUS_FATAL_CRASHES=368,
SERIOUS_INJURIES=406, FATALITIES=11),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='TRACK', TOTAL_CRASHES=35, SERIOUS_FATAL_CRASHES=17,
SERIOUS_INJURIES=18, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='EAST', TOTAL_CRASHES=36, SERIOUS_FATAL_CRASHES=16,
SERIOUS_INJURIES=19, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='PARKWAY', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='LANE',
TOTAL_CRASHES=22, SERIOUS_FATAL_CRASHES=12, SERIOUS_INJURIES=12,
FATALITIES=1),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=326, SERIOUS_FATAL_CRASHES=120,
SERIOUS_INJURIES=127, FATALITIES=2),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=550, SERIOUS_FATAL_CRASHES=238,
SERIOUS_INJURIES=273, FATALITIES=35),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='TRACK', TOTAL_CRASHES=62, SERIOUS_FATAL_CRASHES=24,
SERIOUS_INJURIES=23, FATALITIES=1),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=288, SERIOUS_FATAL_CRASHES=76,
SERIOUS_INJURIES=84, FATALITIES=9),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=159, SERIOUS_FATAL_CRASHES=53,
SERIOUS_INJURIES=64, FATALITIES=1),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=546, SERIOUS_FATAL_CRASHES=292,
SERIOUS_INJURIES=390, FATALITIES=44),
Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='AVENUE',
TOTAL_CRASHES=41, SERIOUS_FATAL_CRASHES=16, SERIOUS_INJURIES=18,
FATALITIES=2),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning',

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ROAD_TYPE='FREEWAY', TOTAL_CRASHES=94, SERIOUS_FATAL_CRASHES=43,
SERIOUS_INJURIES=41, FATALITIES=6),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='WAY', TOTAL_CRASHES=147, SERIOUS_FATAL_CRASHES=43,
SERIOUS_INJURIES=40, FATALITIES=3),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=7, FATALITIES=3),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening', ROAD_TYPE=None,
TOTAL_CRASHES=266, SERIOUS_FATAL_CRASHES=102, SERIOUS_INJURIES=110,
FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='CAUSEWAY',
TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=5, SERIOUS_INJURIES=5,
FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='AVENUE',
TOTAL_CRASHES=15, SERIOUS_FATAL_CRASHES=7, SERIOUS_INJURIES=6,
FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=22, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=1),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon', ROAD_TYPE=None,
TOTAL_CRASHES=39, SERIOUS_FATAL_CRASHES=10, SERIOUS_INJURIES=12,
FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE=None,
TOTAL_CRASHES=30, SERIOUS_FATAL_CRASHES=8, SERIOUS_INJURIES=8,
FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='FREEWAY',
TOTAL_CRASHES=188, SERIOUS_FATAL_CRASHES=87, SERIOUS_INJURIES=99,
FATALITIES=9),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=467, SERIOUS_FATAL_CRASHES=218,
SERIOUS_INJURIES=255, FATALITIES=29),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='ALLEY', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='ROW', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=726, SERIOUS_FATAL_CRASHES=288,
SERIOUS_INJURIES=344, FATALITIES=39),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=18, SERIOUS_FATAL_CRASHES=5,

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SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='EAST',
TOTAL_CRASHES=26, SERIOUS_FATAL_CRASHES=11, SERIOUS_INJURIES=15,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=57, SERIOUS_FATAL_CRASHES=33,
SERIOUS_INJURIES=53, FATALITIES=4),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=520, SERIOUS_FATAL_CRASHES=189,
SERIOUS_INJURIES=205, FATALITIES=6),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='WAY', TOTAL_CRASHES=20, SERIOUS_FATAL_CRASHES=9,
SERIOUS_INJURIES=11, FATALITIES=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='WAY', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=3, FATALITIES=2),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=266, SERIOUS_FATAL_CRASHES=111,
SERIOUS_INJURIES=121, FATALITIES=2),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='RAMP', TOTAL_CRASHES=496, SERIOUS_FATAL_CRASHES=164,
SERIOUS_INJURIES=178, FATALITIES=4),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='PARK',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='CAUSEWAY',
TOTAL_CRASHES=15, SERIOUS_FATAL_CRASHES=4, SERIOUS_INJURIES=5,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='EAST', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=1272, SERIOUS_FATAL_CRASHES=488,
SERIOUS_INJURIES=575, FATALITIES=15),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=1181, SERIOUS_FATAL_CRASHES=392,
SERIOUS_INJURIES=442, FATALITIES=8),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='AVENUE',
TOTAL_CRASHES=54, SERIOUS_FATAL_CRASHES=19, SERIOUS_INJURIES=18,
FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=55, SERIOUS_FATAL_CRASHES=21,
SERIOUS_INJURIES=32, FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='CRESCENT',
TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=5,
FATALITIES=0),

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Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='RAMP', TOTAL_CRASHES=406, SERIOUS_FATAL_CRASHES=143,
SERIOUS_INJURIES=157, FATALITIES=4),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='PLACE', TOTAL_CRASHES=28, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=7, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=615, SERIOUS_FATAL_CRASHES=215,
SERIOUS_INJURIES=231, FATALITIES=4),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='STREET', TOTAL_CRASHES=162, SERIOUS_FATAL_CRASHES=82,
SERIOUS_INJURIES=92, FATALITIES=4),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='PARADE', TOTAL_CRASHES=88, SERIOUS_FATAL_CRASHES=33,
SERIOUS_INJURIES=37, FATALITIES=1),
Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='STREET',
TOTAL_CRASHES=109, SERIOUS_FATAL_CRASHES=42, SERIOUS_INJURIES=50,
FATALITIES=4),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='WEST', TOTAL_CRASHES=18, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=166, SERIOUS_FATAL_CRASHES=66,
SERIOUS_INJURIES=64, FATALITIES=5),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=91, SERIOUS_FATAL_CRASHES=41,
SERIOUS_INJURIES=44, FATALITIES=1),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='TRACK', TOTAL_CRASHES=36, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=8, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='COURT', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=11, FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='COURT', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=4, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight', ROAD_TYPE='WAY',
TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='WAY',
TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=2,
FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening',

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ROAD_TYPE='CRESCENT', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE=None,
TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=2,
FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='STREET', TOTAL_CRASHES=1219, SERIOUS_FATAL_CRASHES=438,
SERIOUS_INJURIES=484, FATALITIES=18),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='WEST', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='ROAD',
TOTAL_CRASHES=3084, SERIOUS_FATAL_CRASHES=1454, SERIOUS_INJURIES=1596,
FATALITIES=154),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=76, SERIOUS_FATAL_CRASHES=47,
SERIOUS_INJURIES=57, FATALITIES=4),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='PARADE', TOTAL_CRASHES=16, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=10, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='RAMP', TOTAL_CRASHES=17, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=11, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PARADE', TOTAL_CRASHES=641, SERIOUS_FATAL_CRASHES=194,
SERIOUS_INJURIES=209, FATALITIES=2),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=47, SERIOUS_FATAL_CRASHES=12,
SERIOUS_INJURIES=14, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='ROAD',
TOTAL_CRASHES=1711, SERIOUS_FATAL_CRASHES=680, SERIOUS_INJURIES=694,
FATALITIES=58),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='RAMP',
TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=3,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PLACE', TOTAL_CRASHES=71, SERIOUS_FATAL_CRASHES=25,
SERIOUS_INJURIES=25, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='RAMP', TOTAL_CRASHES=366, SERIOUS_FATAL_CRASHES=102,
SERIOUS_INJURIES=107, FATALITIES=4),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='WEST', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=6, FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='DRIVE',
TOTAL_CRASHES=79, SERIOUS_FATAL_CRASHES=21, SERIOUS_INJURIES=28,
FATALITIES=2),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='BOULEVARD',
TOTAL_CRASHES=17, SERIOUS_FATAL_CRASHES=4, SERIOUS_INJURIES=4,

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FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='PARKWAY', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='STREET',
TOTAL_CRASHES=1023, SERIOUS_FATAL_CRASHES=287, SERIOUS_INJURIES=320,
FATALITIES=6),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=90, SERIOUS_FATAL_CRASHES=30,
SERIOUS_INJURIES=34, FATALITIES=2),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=20, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=13, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='EAST', TOTAL_CRASHES=15, SERIOUS_FATAL_CRASHES=9,
SERIOUS_INJURIES=10, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='NONE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=2, FATALITIES=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='COURT', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='TRACK', TOTAL_CRASHES=82, SERIOUS_FATAL_CRASHES=43,
SERIOUS_INJURIES=45, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=48, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=15, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='THROUGHWAY', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=33, SERIOUS_FATAL_CRASHES=12,
SERIOUS_INJURIES=13, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='WEST', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),

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Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='ROAD', TOTAL_CRASHES=15611, SERIOUS_FATAL_CRASHES=5564,
SERIOUS_INJURIES=6353, FATALITIES=198),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=1406, SERIOUS_FATAL_CRASHES=463,
SERIOUS_INJURIES=507, FATALITIES=8),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='TRACK', TOTAL_CRASHES=35, SERIOUS_FATAL_CRASHES=14,
SERIOUS_INJURIES=15, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='LANE', TOTAL_CRASHES=29, SERIOUS_FATAL_CRASHES=17,
SERIOUS_INJURIES=17, FATALITIES=3),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='EAST', TOTAL_CRASHES=21, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=7, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=25, SERIOUS_FATAL_CRASHES=13,
SERIOUS_INJURIES=16, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=258, SERIOUS_FATAL_CRASHES=105,
SERIOUS_INJURIES=142, FATALITIES=6),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='PLACE',
TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='STREET', TOTAL_CRASHES=5873, SERIOUS_FATAL_CRASHES=1916,
SERIOUS_INJURIES=2016, FATALITIES=41),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=238, SERIOUS_FATAL_CRASHES=95,
SERIOUS_INJURIES=100, FATALITIES=1),
Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='BOULEVARD',
TOTAL_CRASHES=16, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=5,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='LANE', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=9, FATALITIES=1),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='WYND', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',

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ROAD_TYPE=None, TOTAL_CRASHES=168, SERIOUS_FATAL_CRASHES=72,
SERIOUS_INJURIES=76, FATALITIES=5),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='LANE', TOTAL_CRASHES=64, SERIOUS_FATAL_CRASHES=27,
SERIOUS_INJURIES=26, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='LINK', TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='WEST', TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=8, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='SQUARE', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='RUN', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='STREET', TOTAL_CRASHES=617, SERIOUS_FATAL_CRASHES=163,
SERIOUS_INJURIES=172, FATALITIES=9),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=329, SERIOUS_FATAL_CRASHES=171,
SERIOUS_INJURIES=181, FATALITIES=21),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=434, SERIOUS_FATAL_CRASHES=136,
SERIOUS_INJURIES=143, FATALITIES=4),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='STREET', TOTAL_CRASHES=615, SERIOUS_FATAL_CRASHES=206,
SERIOUS_INJURIES=221, FATALITIES=8),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='RISE', TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='HIGHWAY',
TOTAL_CRASHES=118, SERIOUS_FATAL_CRASHES=70, SERIOUS_INJURIES=70,
FATALITIES=11),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='PARADE', TOTAL_CRASHES=16, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='GROVE', TOTAL_CRASHES=40, SERIOUS_FATAL_CRASHES=14,
SERIOUS_INJURIES=14, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='COURT', TOTAL_CRASHES=136, SERIOUS_FATAL_CRASHES=59,

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SERIOUS_INJURIES=64, FATALITIES=2),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=25, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=17, FATALITIES=2),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='GREEN', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='RAMP', TOTAL_CRASHES=17, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='PLACE', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=133, SERIOUS_FATAL_CRASHES=85,
SERIOUS_INJURIES=87, FATALITIES=9),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='GROVE', TOTAL_CRASHES=28, SERIOUS_FATAL_CRASHES=12,
SERIOUS_INJURIES=13, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='COURT', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=8, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='BREAK', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='ROAD',
TOTAL_CRASHES=436, SERIOUS_FATAL_CRASHES=236, SERIOUS_INJURIES=250,
FATALITIES=34),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='TUNNEL', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='PARADE', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='WAY',
TOTAL_CRASHES=35, SERIOUS_FATAL_CRASHES=9, SERIOUS_INJURIES=10,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='LANE', TOTAL_CRASHES=59, SERIOUS_FATAL_CRASHES=26,
SERIOUS_INJURIES=28, FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='DRIVE',
TOTAL_CRASHES=43, SERIOUS_FATAL_CRASHES=13, SERIOUS_INJURIES=14,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='STREET', TOTAL_CRASHES=1453, SERIOUS_FATAL_CRASHES=374,
SERIOUS_INJURIES=413, FATALITIES=9),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='RETREAT', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',

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ROAD_TYPE='PLACE', TOTAL_CRASHES=29, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=8, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='NONE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=243, SERIOUS_FATAL_CRASHES=135,
SERIOUS_INJURIES=137, FATALITIES=21),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='GROVE', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=51, SERIOUS_FATAL_CRASHES=21,
SERIOUS_INJURIES=25, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='RAMP',
TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='STREET', TOTAL_CRASHES=74, SERIOUS_FATAL_CRASHES=27,
SERIOUS_INJURIES=28, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='BEND', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='LANE',
TOTAL_CRASHES=22, SERIOUS_FATAL_CRASHES=10, SERIOUS_INJURIES=10,
FATALITIES=3),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='RAMP',
TOTAL_CRASHES=43, SERIOUS_FATAL_CRASHES=12, SERIOUS_INJURIES=15,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=753, SERIOUS_FATAL_CRASHES=269,
SERIOUS_INJURIES=296, FATALITIES=6),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='RISE', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=243, SERIOUS_FATAL_CRASHES=125,
SERIOUS_INJURIES=149, FATALITIES=17),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning', ROAD_TYPE='WAY',
TOTAL_CRASHES=151, SERIOUS_FATAL_CRASHES=65, SERIOUS_INJURIES=70,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='WEST', TOTAL_CRASHES=22, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=8, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=114, SERIOUS_FATAL_CRASHES=26,

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SERIOUS_INJURIES=26, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='RAMP', TOTAL_CRASHES=112, SERIOUS_FATAL_CRASHES=46,
SERIOUS_INJURIES=47, FATALITIES=4),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='PLACE', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=209, SERIOUS_FATAL_CRASHES=101,
SERIOUS_INJURIES=107, FATALITIES=5),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=31, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=9, FATALITIES=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='SQUARE', TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=1274, SERIOUS_FATAL_CRASHES=704,
SERIOUS_INJURIES=907, FATALITIES=41),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=3272, SERIOUS_FATAL_CRASHES=1182,
SERIOUS_INJURIES=1390, FATALITIES=40),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='RAMP', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='LANE', TOTAL_CRASHES=21, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=13, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=6, FATALITIES=1),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=27, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='PARADE', TOTAL_CRASHES=99, SERIOUS_FATAL_CRASHES=42,
SERIOUS_INJURIES=48, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='SQUARE', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='ROAD', TOTAL_CRASHES=1228, SERIOUS_FATAL_CRASHES=571,
SERIOUS_INJURIES=622, FATALITIES=56),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='ROAD', TOTAL_CRASHES=1764, SERIOUS_FATAL_CRASHES=862,
SERIOUS_INJURIES=945, FATALITIES=82),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PARADE', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=61,
SERIOUS_INJURIES=67, FATALITIES=2),

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Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION=None, TIME_PERIOD='Morning', ROAD_TYPE=None,
TOTAL_CRASHES=29, SERIOUS_FATAL_CRASHES=14, SERIOUS_INJURIES=14,
FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='BREAK',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ALLEY', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning',
ROAD_TYPE='STREET', TOTAL_CRASHES=930, SERIOUS_FATAL_CRASHES=227,
SERIOUS_INJURIES=228, FATALITIES=10),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='WAY', TOTAL_CRASHES=103, SERIOUS_FATAL_CRASHES=37,
SERIOUS_INJURIES=38, FATALITIES=1),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='DRIVE',
TOTAL_CRASHES=53, SERIOUS_FATAL_CRASHES=19, SERIOUS_INJURIES=21,
FATALITIES=4),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='STREET', TOTAL_CRASHES=526, SERIOUS_FATAL_CRASHES=134,
SERIOUS_INJURIES=139, FATALITIES=5),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='ROAD', TOTAL_CRASHES=1686, SERIOUS_FATAL_CRASHES=765,
SERIOUS_INJURIES=854, FATALITIES=62),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon', ROAD_TYPE='WAY',
TOTAL_CRASHES=74, SERIOUS_FATAL_CRASHES=24, SERIOUS_INJURIES=28,
FATALITIES=3),
Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE=None,
TOTAL_CRASHES=50, SERIOUS_FATAL_CRASHES=26, SERIOUS_INJURIES=23,
FATALITIES=3),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='LINK', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='LANE', TOTAL_CRASHES=32, SERIOUS_FATAL_CRASHES=16,
SERIOUS_INJURIES=19, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=15, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=10, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='CHASE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='STREET', TOTAL_CRASHES=2673, SERIOUS_FATAL_CRASHES=934,
SERIOUS_INJURIES=1000, FATALITIES=13),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',

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ROAD_TYPE='STREET', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=60,
SERIOUS_INJURIES=66, FATALITIES=7),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=217, SERIOUS_FATAL_CRASHES=99,
SERIOUS_INJURIES=121, FATALITIES=6),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='PARADE', TOTAL_CRASHES=353, SERIOUS_FATAL_CRASHES=111,
SERIOUS_INJURIES=122, FATALITIES=1),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=427, SERIOUS_FATAL_CRASHES=225,
SERIOUS_INJURIES=256, FATALITIES=12),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE=None, TOTAL_CRASHES=56, SERIOUS_FATAL_CRASHES=29,
SERIOUS_INJURIES=38, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='TRACK',
TOTAL_CRASHES=143, SERIOUS_FATAL_CRASHES=53, SERIOUS_INJURIES=51,
FATALITIES=4),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='WAY', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=7, FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=894, SERIOUS_FATAL_CRASHES=441,
SERIOUS_INJURIES=483, FATALITIES=44),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=95, SERIOUS_FATAL_CRASHES=43,
SERIOUS_INJURIES=47, FATALITIES=1),
Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='PARADE',
TOTAL_CRASHES=19, SERIOUS_FATAL_CRASHES=6, SERIOUS_INJURIES=6,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=26, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=21, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='COURT', TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=8, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='PARADE', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='ROAD', TOTAL_CRASHES=3075, SERIOUS_FATAL_CRASHES=1528,
SERIOUS_INJURIES=1929, FATALITIES=85),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='PLACE', TOTAL_CRASHES=17, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=4, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='BREAK',
TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=2,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,

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SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='STREET', TOTAL_CRASHES=1379, SERIOUS_FATAL_CRASHES=547,
SERIOUS_INJURIES=615, FATALITIES=6),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=21, SERIOUS_FATAL_CRASHES=10,
SERIOUS_INJURIES=10, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=47, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=15, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE=None,
TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='COURT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='LANE', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='PLACE', TOTAL_CRASHES=47, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=11, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='PARADE', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=23, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=1),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=370, SERIOUS_FATAL_CRASHES=190,
SERIOUS_INJURIES=196, FATALITIES=28),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='DRIVE',
TOTAL_CRASHES=33, SERIOUS_FATAL_CRASHES=10, SERIOUS_INJURIES=12,
FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=45, SERIOUS_FATAL_CRASHES=22,
SERIOUS_INJURIES=27, FATALITIES=2),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='WAY',
TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=6, SERIOUS_INJURIES=7,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='WAY', TOTAL_CRASHES=90, SERIOUS_FATAL_CRASHES=35,
SERIOUS_INJURIES=36, FATALITIES=3),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='WAY', TOTAL_CRASHES=225, SERIOUS_FATAL_CRASHES=69,
SERIOUS_INJURIES=74, FATALITIES=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=3, FATALITIES=0),

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Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='LANE', TOTAL_CRASHES=57, SERIOUS_FATAL_CRASHES=18,
SERIOUS_INJURIES=20, FATALITIES=2),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='VISTA', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=7815, SERIOUS_FATAL_CRASHES=3071,
SERIOUS_INJURIES=3518, FATALITIES=146),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='COURT', TOTAL_CRASHES=64, SERIOUS_FATAL_CRASHES=24,
SERIOUS_INJURIES=24, FATALITIES=1),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=154, SERIOUS_FATAL_CRASHES=103,
SERIOUS_INJURIES=109, FATALITIES=11),
Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='LANE',
TOTAL_CRASHES=35, SERIOUS_FATAL_CRASHES=18, SERIOUS_INJURIES=18,
FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='TRACK', TOTAL_CRASHES=67, SERIOUS_FATAL_CRASHES=30,
SERIOUS_INJURIES=31, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=68, SERIOUS_FATAL_CRASHES=28,
SERIOUS_INJURIES=33, FATALITIES=3),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=87, SERIOUS_FATAL_CRASHES=29,
SERIOUS_INJURIES=42, FATALITIES=1),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PLACE', TOTAL_CRASHES=58, SERIOUS_FATAL_CRASHES=21,
SERIOUS_INJURIES=26, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='BREAK',
TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=8, SERIOUS_INJURIES=8,
FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2063, SERIOUS_FATAL_CRASHES=1070,
SERIOUS_INJURIES=1208, FATALITIES=55),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=386, SERIOUS_FATAL_CRASHES=150,
SERIOUS_INJURIES=164, FATALITIES=6),
Row(DTP_REGION=None, TIME_PERIOD='Afternoon', ROAD_TYPE=None,
TOTAL_CRASHES=30, SERIOUS_FATAL_CRASHES=8, SERIOUS_INJURIES=9,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE=None, TOTAL_CRASHES=27, SERIOUS_FATAL_CRASHES=16,
SERIOUS_INJURIES=19, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',

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ROAD_TYPE='ROAD', TOTAL_CRASHES=2599, SERIOUS_FATAL_CRASHES=1042,
SERIOUS_INJURIES=1152, FATALITIES=80),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='COURT', TOTAL_CRASHES=21, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=9, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='ROAD',
TOTAL_CRASHES=1858, SERIOUS_FATAL_CRASHES=827, SERIOUS_INJURIES=880,
FATALITIES=80),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=130, SERIOUS_FATAL_CRASHES=45,
SERIOUS_INJURIES=52, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='PARADE', TOTAL_CRASHES=30, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=11, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight', ROAD_TYPE=None,
TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=7, SERIOUS_INJURIES=6,
FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='STREET', TOTAL_CRASHES=1491, SERIOUS_FATAL_CRASHES=520,
SERIOUS_INJURIES=563, FATALITIES=9),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=145, SERIOUS_FATAL_CRASHES=53,
SERIOUS_INJURIES=58, FATALITIES=2),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE=None, TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='STREET', TOTAL_CRASHES=10048, SERIOUS_FATAL_CRASHES=3042,
SERIOUS_INJURIES=3170, FATALITIES=65),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='EAST', TOTAL_CRASHES=31, SERIOUS_FATAL_CRASHES=13,
SERIOUS_INJURIES=16, FATALITIES=3),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='STREET', TOTAL_CRASHES=1534, SERIOUS_FATAL_CRASHES=421,
SERIOUS_INJURIES=467, FATALITIES=11),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='RAMP', TOTAL_CRASHES=280, SERIOUS_FATAL_CRASHES=97,
SERIOUS_INJURIES=105, FATALITIES=3),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=31, SERIOUS_FATAL_CRASHES=10,
SERIOUS_INJURIES=13, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='RISE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',
TOTAL_CRASHES=930, SERIOUS_FATAL_CRASHES=414, SERIOUS_INJURIES=452,
FATALITIES=48),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=288, SERIOUS_FATAL_CRASHES=146,

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SERIOUS_INJURIES=149, FATALITIES=14),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='PARKWAY', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='EAST', TOTAL_CRASHES=19, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=13, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='CAUSEWAY',
TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=2,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=100, SERIOUS_FATAL_CRASHES=42,
SERIOUS_INJURIES=40, FATALITIES=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=20, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=7, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='RIDGE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='STREET', TOTAL_CRASHES=137, SERIOUS_FATAL_CRASHES=52,
SERIOUS_INJURIES=69, FATALITIES=7),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='RAMP',
TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='WAY',
TOTAL_CRASHES=61, SERIOUS_FATAL_CRASHES=26, SERIOUS_INJURIES=30,
FATALITIES=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='RAMP', TOTAL_CRASHES=525, SERIOUS_FATAL_CRASHES=179,
SERIOUS_INJURIES=210, FATALITIES=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=23, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=21, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='COURT', TOTAL_CRASHES=78, SERIOUS_FATAL_CRASHES=25,
SERIOUS_INJURIES=25, FATALITIES=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='RISE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='FREEWAY',
TOTAL_CRASHES=132, SERIOUS_FATAL_CRASHES=61, SERIOUS_INJURIES=68,
FATALITIES=8),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=828, SERIOUS_FATAL_CRASHES=391,
SERIOUS_INJURIES=509, FATALITIES=33),

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Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=191, SERIOUS_FATAL_CRASHES=56,
SERIOUS_INJURIES=67, FATALITIES=2),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='STREET', TOTAL_CRASHES=1367, SERIOUS_FATAL_CRASHES=559,
SERIOUS_INJURIES=615, FATALITIES=13),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PROMENADE', TOTAL_CRASHES=20, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=8, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=5, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='LANE', TOTAL_CRASHES=76, SERIOUS_FATAL_CRASHES=18,
SERIOUS_INJURIES=20, FATALITIES=1),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='EAST', TOTAL_CRASHES=43, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=12, FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='LANE', TOTAL_CRASHES=30, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=16, FATALITIES=1),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='TUNNEL', TOTAL_CRASHES=35, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=9, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE=None, TOTAL_CRASHES=74, SERIOUS_FATAL_CRASHES=38,
SERIOUS_INJURIES=40, FATALITIES=1),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='PARADE', TOTAL_CRASHES=17, SERIOUS_FATAL_CRASHES=12,
SERIOUS_INJURIES=15, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='DRIVE',
TOTAL_CRASHES=37, SERIOUS_FATAL_CRASHES=12, SERIOUS_INJURIES=16,
FATALITIES=2),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='TRACK',
TOTAL_CRASHES=30, SERIOUS_FATAL_CRASHES=19, SERIOUS_INJURIES=20,
FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='CRESCENT',
TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=7,
FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE=None, TOTAL_CRASHES=29, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=9, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='PARADE', TOTAL_CRASHES=97, SERIOUS_FATAL_CRASHES=45,
SERIOUS_INJURIES=55, FATALITIES=2),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',

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ROAD_TYPE='SQUARE', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='LINK', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='STREET', TOTAL_CRASHES=567, SERIOUS_FATAL_CRASHES=252,
SERIOUS_INJURIES=275, FATALITIES=8),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='STREET',
TOTAL_CRASHES=395, SERIOUS_FATAL_CRASHES=130, SERIOUS_INJURIES=141,
FATALITIES=6),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='PLACE', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='WAY',
TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=64, SERIOUS_FATAL_CRASHES=26,
SERIOUS_INJURIES=26, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=1369, SERIOUS_FATAL_CRASHES=456,
SERIOUS_INJURIES=522, FATALITIES=8),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=15, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=7, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=15, SERIOUS_FATAL_CRASHES=9,
SERIOUS_INJURIES=12, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='HIGHWAY',
TOTAL_CRASHES=482, SERIOUS_FATAL_CRASHES=244, SERIOUS_INJURIES=273,
FATALITIES=25),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='ROAD', TOTAL_CRASHES=11834, SERIOUS_FATAL_CRASHES=4218,
SERIOUS_INJURIES=4476, FATALITIES=95),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=19, SERIOUS_FATAL_CRASHES=10,
SERIOUS_INJURIES=10, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='COURT',
TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=4,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='GROVE', TOTAL_CRASHES=100, SERIOUS_FATAL_CRASHES=26,
SERIOUS_INJURIES=24, FATALITIES=2),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='RUN',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,

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FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=32, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=11, FATALITIES=1),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=13, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='ACCESS', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=369, SERIOUS_FATAL_CRASHES=187,
SERIOUS_INJURIES=222, FATALITIES=37),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=759, SERIOUS_FATAL_CRASHES=250,
SERIOUS_INJURIES=273, FATALITIES=8),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='CRESCENT',
TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='PATH', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='LANE',
TOTAL_CRASHES=28, SERIOUS_FATAL_CRASHES=9, SERIOUS_INJURIES=10,
FATALITIES=2),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=515, SERIOUS_FATAL_CRASHES=156,
SERIOUS_INJURIES=176, FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='ROUTE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=0,
FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=922, SERIOUS_FATAL_CRASHES=317,
SERIOUS_INJURIES=338, FATALITIES=9),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=449, SERIOUS_FATAL_CRASHES=141,
SERIOUS_INJURIES=157, FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='COURT',
TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='ROAD', TOTAL_CRASHES=10293, SERIOUS_FATAL_CRASHES=3828,
SERIOUS_INJURIES=4210, FATALITIES=148),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='ROAD',

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TOTAL_CRASHES=1104, SERIOUS_FATAL_CRASHES=507, SERIOUS_INJURIES=571,
FATALITIES=42),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='GROVE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE=None, TOTAL_CRASHES=79, SERIOUS_FATAL_CRASHES=37,
SERIOUS_INJURIES=40, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='ROAD', TOTAL_CRASHES=8187, SERIOUS_FATAL_CRASHES=2968,
SERIOUS_INJURIES=3227, FATALITIES=105),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='STREET',
TOTAL_CRASHES=672, SERIOUS_FATAL_CRASHES=199, SERIOUS_INJURIES=204,
FATALITIES=8),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='FREEWAY',
TOTAL_CRASHES=220, SERIOUS_FATAL_CRASHES=107, SERIOUS_INJURIES=126,
FATALITIES=12),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='WEST', TOTAL_CRASHES=15, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=8, FATALITIES=1),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='PARADE', TOTAL_CRASHES=57, SERIOUS_FATAL_CRASHES=23,
SERIOUS_INJURIES=23, FATALITIES=1),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='WEST', TOTAL_CRASHES=15, SERIOUS_FATAL_CRASHES=9,
SERIOUS_INJURIES=12, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='RAMP', TOTAL_CRASHES=90, SERIOUS_FATAL_CRASHES=46,
SERIOUS_INJURIES=49, FATALITIES=4),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='PROMENADE', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=2, FATALITIES=1),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=84, SERIOUS_FATAL_CRASHES=31,
SERIOUS_INJURIES=37, FATALITIES=4),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=284, SERIOUS_FATAL_CRASHES=127,
SERIOUS_INJURIES=143, FATALITIES=7),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='PARADE',
TOTAL_CRASHES=17, SERIOUS_FATAL_CRASHES=5, SERIOUS_INJURIES=5,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='PROMENADE', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=77, SERIOUS_FATAL_CRASHES=29,

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SERIOUS_INJURIES=34, FATALITIES=4),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='PATH', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='LANE', TOTAL_CRASHES=35, SERIOUS_FATAL_CRASHES=14,
SERIOUS_INJURIES=14, FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='LANE',
TOTAL_CRASHES=62, SERIOUS_FATAL_CRASHES=30, SERIOUS_INJURIES=33,
FATALITIES=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=141, SERIOUS_FATAL_CRASHES=74,
SERIOUS_INJURIES=72, FATALITIES=17),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=451, SERIOUS_FATAL_CRASHES=173,
SERIOUS_INJURIES=188, FATALITIES=3),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=113, SERIOUS_FATAL_CRASHES=53,
SERIOUS_INJURIES=51, FATALITIES=11),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='RAMP',
TOTAL_CRASHES=35, SERIOUS_FATAL_CRASHES=12, SERIOUS_INJURIES=13,
FATALITIES=2),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='LANE', TOTAL_CRASHES=45, SERIOUS_FATAL_CRASHES=16,
SERIOUS_INJURIES=17, FATALITIES=3),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='FREEWAY',
TOTAL_CRASHES=306, SERIOUS_FATAL_CRASHES=159, SERIOUS_INJURIES=208,
FATALITIES=21),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='WEST', TOTAL_CRASHES=24, SERIOUS_FATAL_CRASHES=17,
SERIOUS_INJURIES=19, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='TRACK', TOTAL_CRASHES=53, SERIOUS_FATAL_CRASHES=18,
SERIOUS_INJURIES=19, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=19, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=12, FATALITIES=1),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='PARADE',
TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE=None,
TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=69, SERIOUS_FATAL_CRASHES=32,
SERIOUS_INJURIES=34, FATALITIES=0),

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Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='ROW', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='STREET', TOTAL_CRASHES=947, SERIOUS_FATAL_CRASHES=383,
SERIOUS_INJURIES=419, FATALITIES=11),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='LANE', TOTAL_CRASHES=75, SERIOUS_FATAL_CRASHES=31,
SERIOUS_INJURIES=39, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='TRACK',
TOTAL_CRASHES=18, SERIOUS_FATAL_CRASHES=10, SERIOUS_INJURIES=10,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2051, SERIOUS_FATAL_CRASHES=1045,
SERIOUS_INJURIES=1227, FATALITIES=58),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=849, SERIOUS_FATAL_CRASHES=324,
SERIOUS_INJURIES=354, FATALITIES=12),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='COURT', TOTAL_CRASHES=94, SERIOUS_FATAL_CRASHES=41,
SERIOUS_INJURIES=44, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='DRIVE',
TOTAL_CRASHES=73, SERIOUS_FATAL_CRASHES=27, SERIOUS_INJURIES=28,
FATALITIES=1),
Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='COURT',
TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=6, SERIOUS_INJURIES=6,
FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='LANE',
TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=5, SERIOUS_INJURIES=6,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=6, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='MEWS', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE=None,
TOTAL_CRASHES=18, SERIOUS_FATAL_CRASHES=9, SERIOUS_INJURIES=11,
FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=130, SERIOUS_FATAL_CRASHES=50,
SERIOUS_INJURIES=52, FATALITIES=1),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='RAMP', TOTAL_CRASHES=309, SERIOUS_FATAL_CRASHES=110,
SERIOUS_INJURIES=126, FATALITIES=2),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',

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ROAD_TYPE='PLACE', TOTAL_CRASHES=49, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=16, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=33, SERIOUS_FATAL_CRASHES=13,
SERIOUS_INJURIES=13, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='AVENUE',
TOTAL_CRASHES=62, SERIOUS_FATAL_CRASHES=16, SERIOUS_INJURIES=17,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=50, SERIOUS_FATAL_CRASHES=13,
SERIOUS_INJURIES=13, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='RAMP', TOTAL_CRASHES=34, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=18, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=389, SERIOUS_FATAL_CRASHES=218,
SERIOUS_INJURIES=281, FATALITIES=20),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=37, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=17, FATALITIES=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='TRACK', TOTAL_CRASHES=13, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=11, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='TRACK',
TOTAL_CRASHES=31, SERIOUS_FATAL_CRASHES=12, SERIOUS_INJURIES=11,
FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE=None, TOTAL_CRASHES=285, SERIOUS_FATAL_CRASHES=108,
SERIOUS_INJURIES=122, FATALITIES=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=636, SERIOUS_FATAL_CRASHES=268,
SERIOUS_INJURIES=322, FATALITIES=12),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='PARADE', TOTAL_CRASHES=465, SERIOUS_FATAL_CRASHES=159,
SERIOUS_INJURIES=170, FATALITIES=5),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='PLAZA', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE=None, TOTAL_CRASHES=41, SERIOUS_FATAL_CRASHES=18,
SERIOUS_INJURIES=19, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='ROAD', TOTAL_CRASHES=15844, SERIOUS_FATAL_CRASHES=5214,

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SERIOUS_INJURIES=5664, FATALITIES=115),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='WAY', TOTAL_CRASHES=57, SERIOUS_FATAL_CRASHES=20,
SERIOUS_INJURIES=25, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning', ROAD_TYPE=None,
TOTAL_CRASHES=429, SERIOUS_FATAL_CRASHES=132, SERIOUS_INJURIES=144,
FATALITIES=4),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE=None, TOTAL_CRASHES=44, SERIOUS_FATAL_CRASHES=20,
SERIOUS_INJURIES=23, FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='STREET', TOTAL_CRASHES=266, SERIOUS_FATAL_CRASHES=120,
SERIOUS_INJURIES=144, FATALITIES=6),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=73, SERIOUS_FATAL_CRASHES=33,
SERIOUS_INJURIES=35, FATALITIES=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='RISE', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='SQUARE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=921, SERIOUS_FATAL_CRASHES=305,
SERIOUS_INJURIES=325, FATALITIES=9),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ROAD', TOTAL_CRASHES=2240, SERIOUS_FATAL_CRASHES=1105,
SERIOUS_INJURIES=1229, FATALITIES=104),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=660, SERIOUS_FATAL_CRASHES=312,
SERIOUS_INJURIES=387, FATALITIES=51),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=126, SERIOUS_FATAL_CRASHES=69,
SERIOUS_INJURIES=76, FATALITIES=11),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='GROVE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PARKWAY', TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=6, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=36, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=17, FATALITIES=1),

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Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='STREET', TOTAL_CRASHES=7689, SERIOUS_FATAL_CRASHES=2491,
SERIOUS_INJURIES=2549, FATALITIES=46),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='WALK', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='COURT', TOTAL_CRASHES=24, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=11, FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='PARADE', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=23, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=8, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=577, SERIOUS_FATAL_CRASHES=277,
SERIOUS_INJURIES=338, FATALITIES=22),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='TRACK', TOTAL_CRASHES=248, SERIOUS_FATAL_CRASHES=79,
SERIOUS_INJURIES=83, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=32, SERIOUS_FATAL_CRASHES=17,
SERIOUS_INJURIES=22, FATALITIES=1),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=138, SERIOUS_FATAL_CRASHES=68,
SERIOUS_INJURIES=84, FATALITIES=7),
Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='WAY',
TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=4, SERIOUS_INJURIES=5,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='RAMP', TOTAL_CRASHES=20, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=7, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='RAMP',
TOTAL_CRASHES=20, SERIOUS_FATAL_CRASHES=5, SERIOUS_INJURIES=4,
FATALITIES=1),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='WEST', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='LANE', TOTAL_CRASHES=23, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=12, FATALITIES=1),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='GROVE', TOTAL_CRASHES=42, SERIOUS_FATAL_CRASHES=14,
SERIOUS_INJURIES=13, FATALITIES=1),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=147, SERIOUS_FATAL_CRASHES=47,
SERIOUS_INJURIES=52, FATALITIES=4),
Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='WAY',

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TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=4, SERIOUS_INJURIES=5,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='LINK', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=90, SERIOUS_FATAL_CRASHES=33,
SERIOUS_INJURIES=34, FATALITIES=2),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=46, SERIOUS_FATAL_CRASHES=24,
SERIOUS_INJURIES=29, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=704, SERIOUS_FATAL_CRASHES=268,
SERIOUS_INJURIES=335, FATALITIES=6),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='COURT', TOTAL_CRASHES=29, SERIOUS_FATAL_CRASHES=8,
SERIOUS_INJURIES=8, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='COURT', TOTAL_CRASHES=17, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=6, FATALITIES=1),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='STREET', TOTAL_CRASHES=872, SERIOUS_FATAL_CRASHES=208,
SERIOUS_INJURIES=219, FATALITIES=8),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE=None, TOTAL_CRASHES=23, SERIOUS_FATAL_CRASHES=10,
SERIOUS_INJURIES=10, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='TUNNEL', TOTAL_CRASHES=36, SERIOUS_FATAL_CRASHES=9,
SERIOUS_INJURIES=10, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE=None, TOTAL_CRASHES=359, SERIOUS_FATAL_CRASHES=145,
SERIOUS_INJURIES=161, FATALITIES=1),
  Row(DTP_REGION=None, TIME_PERIOD='Overnight', ROAD_TYPE=None,
TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=2,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=513, SERIOUS_FATAL_CRASHES=201,
SERIOUS_INJURIES=214, FATALITIES=14),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=69, SERIOUS_FATAL_CRASHES=26,
SERIOUS_INJURIES=29, FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='NONE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='DRIVE',
TOTAL_CRASHES=53, SERIOUS_FATAL_CRASHES=11, SERIOUS_INJURIES=10,

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FATALITIES=1),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening',
ROAD_TYPE='PARADE', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='ROAD',
TOTAL_CRASHES=996, SERIOUS_FATAL_CRASHES=486, SERIOUS_INJURIES=553,
FATALITIES=61),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=62, SERIOUS_FATAL_CRASHES=18,
SERIOUS_INJURIES=19, FATALITIES=2),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='STREET', TOTAL_CRASHES=985, SERIOUS_FATAL_CRASHES=275,
SERIOUS_INJURIES=288, FATALITIES=10),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='LANE', TOTAL_CRASHES=110, SERIOUS_FATAL_CRASHES=28,
SERIOUS_INJURIES=32, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='LANE',
TOTAL_CRASHES=39, SERIOUS_FATAL_CRASHES=15, SERIOUS_INJURIES=15,
FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='CRESCENT',
TOTAL_CRASHES=13, SERIOUS_FATAL_CRASHES=6, SERIOUS_INJURIES=6,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=40, SERIOUS_FATAL_CRASHES=11,
SERIOUS_INJURIES=13, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=2095, SERIOUS_FATAL_CRASHES=792,
SERIOUS_INJURIES=877, FATALITIES=28),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=66, SERIOUS_FATAL_CRASHES=42,
SERIOUS_INJURIES=50, FATALITIES=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE=None, TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=133, SERIOUS_FATAL_CRASHES=55,
SERIOUS_INJURIES=59, FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CHASE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=127, SERIOUS_FATAL_CRASHES=49,
SERIOUS_INJURIES=53, FATALITIES=5),

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Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='CAUSEWAY',
TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning',
ROAD_TYPE='PARADE', TOTAL_CRASHES=18, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='TUNNEL', TOTAL_CRASHES=12, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=366, SERIOUS_FATAL_CRASHES=118,
SERIOUS_INJURIES=131, FATALITIES=3),
Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='TRACK',
TOTAL_CRASHES=237, SERIOUS_FATAL_CRASHES=94, SERIOUS_INJURIES=92,
FATALITIES=3),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='LANE', TOTAL_CRASHES=135, SERIOUS_FATAL_CRASHES=47,
SERIOUS_INJURIES=47, FATALITIES=3),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=41, SERIOUS_FATAL_CRASHES=14,
SERIOUS_INJURIES=16, FATALITIES=1),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon', ROAD_TYPE='WAY',
TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=2,
FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='CIRCUIT',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=2,
FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='GROVE', TOTAL_CRASHES=66, SERIOUS_FATAL_CRASHES=18,
SERIOUS_INJURIES=19, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='HIGHWAY',
TOTAL_CRASHES=323, SERIOUS_FATAL_CRASHES=167, SERIOUS_INJURIES=192,
FATALITIES=25),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=24, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=8, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=165, SERIOUS_FATAL_CRASHES=46,
SERIOUS_INJURIES=47, FATALITIES=1),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=94, SERIOUS_FATAL_CRASHES=43,
SERIOUS_INJURIES=48, FATALITIES=2),
Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='TRAIL',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='FREEWAY', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=10, FATALITIES=6),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',

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ROAD_TYPE='WAY', TOTAL_CRASHES=16, SERIOUS_FATAL_CRASHES=6,
SERIOUS_INJURIES=6, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='COURT', TOTAL_CRASHES=41, SERIOUS_FATAL_CRASHES=17,
SERIOUS_INJURIES=19, FATALITIES=1),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=5, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='PARKWAY', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=5, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='PARADE', TOTAL_CRASHES=87, SERIOUS_FATAL_CRASHES=36,
SERIOUS_INJURIES=38, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='CLOSE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='WAY', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=4, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon', ROAD_TYPE=None,
TOTAL_CRASHES=23, SERIOUS_FATAL_CRASHES=6, SERIOUS_INJURIES=5,
FATALITIES=1),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='HIGHWAY', TOTAL_CRASHES=267, SERIOUS_FATAL_CRASHES=121,
SERIOUS_INJURIES=149, FATALITIES=17),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='EAST', TOTAL_CRASHES=33, SERIOUS_FATAL_CRASHES=15,
SERIOUS_INJURIES=17, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='LANE', TOTAL_CRASHES=76, SERIOUS_FATAL_CRASHES=26,
SERIOUS_INJURIES=29, FATALITIES=1),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening', ROAD_TYPE='WAY',
TOTAL_CRASHES=157, SERIOUS_FATAL_CRASHES=68, SERIOUS_INJURIES=73,
FATALITIES=2),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='DRIVE', TOTAL_CRASHES=47, SERIOUS_FATAL_CRASHES=17,
SERIOUS_INJURIES=20, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='PARADE', TOTAL_CRASHES=136, SERIOUS_FATAL_CRASHES=46,
SERIOUS_INJURIES=50, FATALITIES=2),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='COURT', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=6, FATALITIES=1),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='SQUARE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='TRACK',
TOTAL_CRASHES=148, SERIOUS_FATAL_CRASHES=52, SERIOUS_INJURIES=54,

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FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='PROMENADE', TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='WYND', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='BREAK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='QUAY', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='PARADE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='PLACE', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='RUN', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='RAMP', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='RAMP', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='BEND', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PASS', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='CLOSE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',

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ROAD_TYPE='PROMENADE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='PLAZA',
TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='RISE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='RISE', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='TRAIL', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='COURT',
TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='BOULEVARDE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=0, FATALITIES=1),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='GROVE', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PATH', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='WAY', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=7,
SERIOUS_INJURIES=7, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='COURT', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='BREAK',
TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='STRIP', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=0,

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SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='QUADRANT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='NONE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='GREEN',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='TRACK', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='EAST',
TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=5,
FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='PLACE',
TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=2,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='HUB', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='WAY', TOTAL_CRASHES=13, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=7, FATALITIES=2),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='EAST', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=2, FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='WALK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='COURT', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=5, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='FIRELINE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),

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Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='SQUARE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='VISTA', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE=None,
TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=5, SERIOUS_INJURIES=5,
FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CUTTING', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='WEST', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=6, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='TRACK', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=6, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='RISE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='GROVE', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=5, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='RAMP', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='GLEN', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='PLACE', TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=2, FATALITIES=1),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='COURT',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=0,
FATALITIES=1),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='TRACK', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='TRACK',

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TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='TRACK', TOTAL_CRASHES=10, SERIOUS_FATAL_CRASHES=5,
SERIOUS_INJURIES=6, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='BOULEVARDE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='CIRCUIT',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=2, SERIOUS_INJURIES=2,
FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='PARADE',
TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=5, SERIOUS_INJURIES=5,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='PLAZA', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='PLACE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='PARADE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='LINK', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='AVENUE', TOTAL_CRASHES=11, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=1),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='TRAIL', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='ENTRANCE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='ALLEY', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='PATH', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='EAST', TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3,

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SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='GROVE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='PROMENADE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='PARKWAY', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='W',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=2,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='PLAZA',
TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='RISE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='CLOSE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='WALK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='TRACK', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='COURT', TOTAL_CRASHES=9, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='RISE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='TRACK', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='COURT',
TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),

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Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='WAY', TOTAL_CRASHES=7, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='PLACE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='WEST', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon', ROAD_TYPE='W',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=0,
FATALITIES=1),
Row(DTP_REGION='METROPOLITAN SOUTH EAST REGION',
TIME_PERIOD='Afternoon', ROAD_TYPE=None, TOTAL_CRASHES=1,
SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='TRAIL',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CONCOURSE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='BEND', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='TERRACE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='POINT', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='RIDGE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='PLACE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening',
ROAD_TYPE='SQUARE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='RAMP', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='GROVE', TOTAL_CRASHES=5, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=5, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='COURT',

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TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='PLACE',
TOTAL_CRASHES=8, SERIOUS_FATAL_CRASHES=3, SERIOUS_INJURIES=3,
FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='GROVE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='BREAK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='WALK', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='BREAK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='ROW', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='BREAK',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARDE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening', ROAD_TYPE='CLOSE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='GRANGE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='GROVE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='FIRETRAIL',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='E',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='LOOP', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='NOOK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,

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SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='MEWS', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='GROVE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=4,
FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='ALLEY', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='BOULEVARDE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='NONE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='CIRCUIT',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='CONCOURSE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='COURT', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='ALT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='ESPLANADE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='RISE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),

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Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=14, SERIOUS_FATAL_CRASHES=4,
SERIOUS_INJURIES=7, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CONCOURSE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='GLADE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='RISE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='TERRACE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='STREET
S', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='TRACK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='FIREBREAK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='FIRELINE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='CHASE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='NONE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='WALK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='TRAIL', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='RAMP', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='HUB', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',

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ROAD_TYPE='LANE', TOTAL_CRASHES=6, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=4, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='MEWS', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning',
ROAD_TYPE='CRESCENT', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='RETREAT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=1),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='GROVE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Evening',
ROAD_TYPE='CIRCUIT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='RAMP', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning', ROAD_TYPE='PLACE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=0,
FATALITIES=1),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='S', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CENTRE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='QUADRANT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='PARK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Morning',
ROAD_TYPE='STRIP', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='PARADE', TOTAL_CRASHES=4, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='RUN', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,

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SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='GREEN', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning',
ROAD_TYPE='FIRELINE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='WYND', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='WYND', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='GREEN', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='FIRELINE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Overnight',
ROAD_TYPE='BOULEVARD', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='GRANGE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='CONCOURSE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Morning', ROAD_TYPE='GROVE',
TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='WALK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Evening',
ROAD_TYPE='CLOSE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Afternoon', ROAD_TYPE='CIRCUIT',
TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Morning', ROAD_TYPE='MEWS',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='VALLEY', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='SQUARE',

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TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='GARDENS', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=3,
SERIOUS_INJURIES=3, FATALITIES=0),
  Row(DTP_REGION='METROPOLITAN SOUTH EAST REGION',
TIME_PERIOD='Morning', ROAD_TYPE=None, TOTAL_CRASHES=1,
SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARDE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening',
ROAD_TYPE='GARDENS', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Morning',
ROAD_TYPE='BOULEVARDE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=0, FATALITIES=1),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='GATEWAY', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='RISE', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='LANE', TOTAL_CRASHES=3, SERIOUS_FATAL_CRASHES=2,
SERIOUS_INJURIES=2, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='BEND', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='SQUARE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='PROMENADE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='CIRCLE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='WALK', TOTAL_CRASHES=2, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='RIDGE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='TRAIL', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,

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SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Afternoon',
ROAD_TYPE='EAST', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Overnight',
ROAD_TYPE='TERRACE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='QUADRANT', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='PLACE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='COURT',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='TRAIL', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='CENTRE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Afternoon', ROAD_TYPE='E',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='BEND', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
  Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='GROVE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='VISTA', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='HUME', TIME_PERIOD='Overnight', ROAD_TYPE='WAY',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
  Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='GREEN', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='LODDON MALLEE', TIME_PERIOD='Morning',
ROAD_TYPE='RISE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='VISTA', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
  Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='CLOSE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),

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Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='LINK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='HUME', TIME_PERIOD='Evening', ROAD_TYPE='PLACE',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0, SERIOUS_INJURIES=0,
FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='LOOP', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Overnight',
ROAD_TYPE='STRIP', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='WALK', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Evening', ROAD_TYPE='TRACK',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='VISTA', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Afternoon',
ROAD_TYPE='BYPASS', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=3, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='GRANGE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Evening',
ROAD_TYPE='TRAIL', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='CIRCLE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='BARWON SOUTH WEST', TIME_PERIOD='Evening',
ROAD_TYPE='THROUGHWAY', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GREATER METRO', TIME_PERIOD='Afternoon',
ROAD_TYPE='BOULEVARDE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='INNER METRO', TIME_PERIOD='Morning',
ROAD_TYPE='COURSE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=0,
SERIOUS_INJURIES=0, FATALITIES=0),
Row(DTP_REGION='GIPPSLAND', TIME_PERIOD='Overnight',
ROAD_TYPE='PARADE', TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1,
SERIOUS_INJURIES=1, FATALITIES=0),
Row(DTP_REGION='GRAMPIANS', TIME_PERIOD='Overnight', ROAD_TYPE='WAY',
TOTAL_CRASHES=1, SERIOUS_FATAL_CRASHES=1, SERIOUS_INJURIES=1,
FATALITIES=0)]

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DAG Screenshot

The DAG shows multiple execution stages, with separation occurring around the two Exchange operators. This represents the shuffle operations previously identified in the physical execution plan. The first one occurs after a partial HashAggregate where Spark redistributed the data according to the grouping columns and ACCIDENT_NO to get a distinct crash count. The second shuffle moves the partially aggregated data using DTP_REGION, TIME_PERIOD, and ROAD_TYPE so that the final group level aggregation can be processed.

The DAG photo also shows AQEShuffleRead and AdaptiveSparkPlan. In a distributed cluster, the shuffle could require intermediate data to be transferred between executors, which increases network I/O and scheduling overhead. Since the application is being run locally, the shuffle occurs within the local machine instead of between physical cluster nodes.

No data skew is obvious in the DAG visualisation, which is consistent with earlier hash-partitioning that showed relatively evenly distributed volumes across the partitions.

Potential optimisation could be in the form of reducing columns prior to aggregation, filtering data as early as possible so that partial aggregation can reduce shuffle volume. Pre-partitioning on frequently used columns could reduce redistribution for repeated workloads.

